

# Synthesizing the Consensus: Generative AI and the Pricing of Multi-Provider Earnings Forecasts\*

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## Abstract

The consensus earnings forecast is not one number: forecast data providers (FDPs) construct economically distinct consensus numbers for the same firm-quarter. I show that generative AI (GenAI) changes how equity markets synthesize consensus forecasts across providers. In a panel of S&P 1500 firm-quarters from 2021 to 2025 covered by all five major FDPs, announcement reactions after ChatGPT's release align more closely with provider accuracy ranks and with the consensus's predictive value. Price reactions also shift from the I/B/E/S-only and equal-weighted heuristics toward an accuracy-weighted consensus across the five FDPs. A long-short strategy trading the gap between the accuracy-weighted and heuristic consensus measures earns 0.63–1.35 percent per announcement. The advantage of the accuracy-weighted consensus over I/B/E/S tracks the retrieval capabilities of the underlying large language models (LLMs): negative before the models could browse, positive under browsing, and significantly positive only under native search. The shift concentrates where GenAI-enabled synthesis is most valuable—in firm-quarters with high cross-FDP disagreement and large nonrecurring items—and, once the models can browse, in firms whose forecasts they demonstrably retrieve (high LLM visibility). The evidence suggests that GenAI helps investors both gather and integrate forecast information—and changes how that information is impounded into prices.

**Keywords:** Generative AI, Analyst, Forecast Data Provider, Consensus Forecast, Earnings Surprise, Information Processing, Earnings Response Coefficient.

**JEL classification:** G11, G14, G41, M41.

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## 1 Introduction

“The consensus” earnings forecast is not one number. Sell-side analyst forecasts are the market’s primary benchmark for evaluating realized earnings (e.g., [Bradshaw et al., 2017](#); [Kothari, 2001](#)). The earnings surprise relative to this consensus is a key determinant of announcement-window stock price reactions and post-announcement drift (e.g., [Bartov et al., 2002](#); [Bernard and Thomas, 1989](#); [Livnat and Mendenhall, 2006](#)), with measurable consequences for price discovery, trading volume, and firms’ own disclosure choices. Yet multiple *forecast data providers* (FDPs) construct economically distinct consensus numbers for the same firm-quarter through proprietary aggregation rules. [Larocque et al. \(2025\)](#) document these differences across the five leading FDPs: I/B/E/S, FactSet, Bloomberg, S&P Capital IQ, and Zacks.<sup>1</sup> Although these FDPs are widely used and cited for investment decisions ([Coleman et al., 2025](#); [Xue et al., 2025](#)), little is known about whether the market discriminates among FDPs by their accuracy and, more importantly, how it integrates across them.

The question matters because acquiring and integrating data across multiple providers is costly.<sup>2</sup> Since the release of ChatGPT, generative AI (GenAI) has diffused rapidly into investor research and analysis workflows, reshaping the economics of how financial information is gathered and combined. Industry surveys document widespread adoption ([CFA Institute, 2024](#)), and recent research shows that these tools substantially lower the costs of acquiring and integrating financial information (e.g., [Bertomeu et al., 2025](#); [Blankespoor et al., 2026](#); [Chang et al., 2026](#); [Cheng et al., 2025](#); [Liu and Subasi, 2026](#); [Xue et al., 2025](#)). Prior technological advances such as EDGAR, XBRL, and social-media disclosure each lowered a specific component of investors’ information-processing costs (e.g., [Blankespoor, 2019](#); [Blankespoor et al., 2014](#); [Drake et al., 2015](#)), but none automated what GenAI uniquely enables: cross-FDP synthesis of analyst consensus forecasts. This paper asks whether GenAI has shifted the market’s implicit consensus toward an accuracy-weighted

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<sup>1</sup>In a sample of 94,030 firm-quarters covered by all five FDPs over 2002–2020, [Larocque et al. \(2025\)](#) find that the FDPs report different earnings surprises (beyond rounding) for the same firm-quarter 58 percent of the time, and that at least one FDP signals a miss while another signals a beat in 9 percent of firm-quarters.

<sup>2</sup>[Larocque et al. \(2025\)](#) show that announcement-window returns align more with I/B/E/S and Capital IQ signals than with the other FDPs, consistent with acquisition cost being a binding factor.

multi-FDP synthesis and away from the conventional single-FDP consensus or the equal-weighted average across providers.<sup>3</sup> It has: from pre- to post-ChatGPT, the announcement return generated by a one-percentage-point standardized surprise rises 2.67 percentage points more under the accuracy-weighted consensus than under I/B/E/S, and a quintile long–short portfolio sorted on the gap between the two consensus numbers earns 1.35 percent per announcement.

The case for cross-FDP synthesis rests on the substantive disagreement among providers, which differ in their screening rules, treatment of unusual items, and analyst-coverage thresholds (Bochkay et al., 2022). Which FDP yields the most accurate consensus varies across firms and over time; I/B/E/S ranks highest on average, consistent with its status as the academic-research standard (Larocque et al., 2025). This variation implies that synthesizing across providers can improve on any fixed choice of provider. Historically, such synthesis was impractical: investors had to recognize that the FDPs disagree, subscribe to each consensus separately, and reconcile differing methodologies.

Recognize, subscribe, reconcile—these three frictions correspond to the awareness, acquisition, and integration costs in the disclosure-processing-cost framework of Blankespoor et al. (2020). GenAI attenuates all three, but the binding friction for accuracy-weighted multi-FDP aggregation has historically been the third: cross-source integration of heterogeneous consensus measures, which prior single-source technologies left intact.<sup>4</sup> The framework yields two primary empirical predictions: (i) the post-ChatGPT market should shift its implicit consensus toward an accuracy-weighted synthesis across multiple FDPs, and (ii) this shift should concentrate where GenAI synthesis is most accessible and most valuable.

Neither prediction is obvious ex ante. The strongest reason for doubt is theoretical: in the forecast-combination literature, combining heterogeneous, partially independent forecasts almost

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<sup>3</sup>The accuracy-weighted construction is not a claim that the market literally applies this weighting rule. It serves as a tractable proxy for the broader hypothesis that the post-ChatGPT market draws on multiple FDPs and discriminates by source accuracy; if so, market reactions should align more closely with this benchmark than with the single-FDP or equal-weighted alternatives, even if the specific weights the market applies differ from mine.

<sup>4</sup>A single large language model (LLM) query of the form “What is the consensus EPS forecast for [ticker] next quarter?” typically returns a synthesized response citing several free third-party distributors (e.g., Yahoo Finance, MarketBeat, TipRanks). Appendix B reports two worked examples (IBM and Microsoft) showing the cited distributors, their reported figures, and the disclosed upstream FDP where available.

always dominates sub-selecting on accuracy (e.g., [Ahlers and Lakonishok, 1983](#); [Bates and Granger, 1969](#)), and estimated combination weights rarely beat equal weights out of sample ([Clement, 1999](#); [Timmermann, 2006](#)). Consistent with this logic, [Michael et al. \(2024\)](#) show that constructing a consensus from high-quality analysts within a single provider does not improve investor outcomes; the frictions they identify carry over with even greater force to the cross-FDP setting. A separate concern is that lower acquisition costs do not imply more selective integration: a richer information set need not yield a more accurate consensus when the synthesis itself is unselective, and LLMs in particular can rely heavily on memorization rather than reasoning (e.g., [Lopez-Lira et al., 2025](#)). Whether GenAI nonetheless shifts the market toward a more selective synthesis is therefore an empirical question.

I test these predictions using a panel of S&P 1500 firm-quarters from 2021 to 2025 with consensus EPS forecasts for the same firm-quarter from all five FDPs.<sup>5</sup> After ChatGPT’s release, the market reacts more strongly to forecasts from more accurate FDPs and discounts those from less accurate ones. In a within-firm-quarter rank-based test that estimates provider-specific earnings response coefficients (ERCs; e.g., [Collins and Kothari, 1989](#); [Easton and Zmijewski, 1989](#)) by accuracy rank, the ERC increment per one-rank accuracy gain roughly doubles after ChatGPT, from 0.306 to 0.621 (post-pre differential 0.316,  $p < 0.01$ ). A complementary firm-level design partitions firms by whether I/B/E/S’s accuracy rank rose or fell from pre- to post-ChatGPT. The result is one-sided: the market response to I/B/E/S-based earnings surprises weakens significantly where I/B/E/S lost rank, while the corresponding estimate where it gained is smaller and statistically indistinguishable from zero, consistent with pre-GenAI overweighting of the I/B/E/S consensus. The same selectivity appears on a second quality dimension: when I score each firm’s consensus in real time by how closely it has anticipated the year-ahead street numbers, the ERC gradient on this *predictive value* roughly doubles after ChatGPT.

Given this evidence of selective use, I next test whether the market’s aggregation rule itself

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<sup>5</sup>The sample begins in 2021 for two reasons: (i) it yields a relatively balanced panel of pre- and post-ChatGPT firm-quarters around the November 30, 2022 cut, and (ii) Bloomberg’s historical consensus data are accessible only for the past five years. Given the different methodologies used by FDPs, I compute each FDP’s forecast accuracy from its own end-of-quarter consensus EPS and reported actual EPS.

has shifted toward an accuracy-weighted synthesis. I construct an accuracy-weighted standardized unexpected earnings (SUE) measure in which each FDP’s SUE is weighted by its trailing forecast accuracy, and benchmark it against two heuristic alternatives: the single-FDP I/B/E/S consensus and the equal-weighted average across the five FDPs.<sup>6</sup> Post-ChatGPT, ERCs on the accuracy-weighted surprise rise sharply, while ERCs on the I/B/E/S and equal-weighted surprises remain flat or decline. The resulting gap in ERCs—the *accuracy premium*—widens significantly against both heuristic benchmarks. In economic terms, the post-ChatGPT increments in the three-day announcement cumulative abnormal return (CAR) per percentage point of standardized surprise—2.40 percentage points against the equal-weighted benchmark and 2.67 against I/B/E/S—are each about the size of the entire pre-ChatGPT ERC on the I/B/E/S consensus itself: the market has added to the accuracy-weighted synthesis roughly the weight it once gave I/B/E/S.

The timing and the mechanism of this shift both point to GenAI. Re-estimated within four GPT-capability regimes, the premium against I/B/E/S climbs from negative in the pre-ChatGPT and text-only regimes to 1.52 percentage points under browsing and a significant 4.91 under native search—a staircase in point estimates that tracks the tools’ retrieval capabilities. Integration operates only on the forecasts a model can find, so acquisition gates the premium. A firm-level *LLM visibility* measure—how many forecast sources a browsing-enabled model actually retrieves for the firm—is empirically distinct from classical prominence proxies,<sup>7</sup> and the premium concentrates in high-visibility firms only once the model can browse. Together, these shifts imply that the consensus reflected in market prices has moved from heuristic alternatives toward an accuracy-weighted multi-FDP synthesis.

Two cross-sectional tests show that the accuracy premium concentrates in settings where GenAI-enabled synthesis is most valuable. Because the regime evidence localizes the premium to the browsing and native-search regimes, these tests date the shift from the onset of GPT browsing

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<sup>6</sup>Specifically, the weight on FDP  $g$  is the softmax of the negative of its  $z$ -scored four-quarter rolling mean absolute forecast error, so providers with better recent track records receive larger weights. [Figure 2](#) confirms that the accuracy-weighted SUE has the smallest mean absolute forecast error of the three consensus measures, with the difference statistically significant. All variable definitions are in [Appendix A](#).

<sup>7</sup>Operationally, LLM visibility is an EPS-extraction indicator times  $\log(1 + \text{unique cited domains})$ , averaged across ChatGPT and Perplexity responses to firm-specific EPS queries (from the DataForSEO LLM Responses API).

(September 27, 2023), when web retrieval first made cross-provider synthesis feasible, rather than from ChatGPT’s release. First, the premium concentrates in firm-quarters with high cross-FDP disagreement, where accuracy-weighting offers the largest improvement over heuristic benchmarks. Second, it concentrates in firm-quarters with large special items: because FDPs differ most in their treatment of such nonrecurring items (Bochkay et al., 2022), this is where synthesizing across providers does the most to offset any single provider’s methodology choices.

The accuracy-weighted consensus is also worth trading on.<sup>8</sup> I construct a predicted earnings surprise—the price-scaled difference between the accuracy-weighted and each heuristic consensus—as an ex ante predictor of announcement-window returns. Following the earnings-surprise trading-strategy literature (e.g., Battalio and Mendenhall, 2005; Michaely et al., 2024), I form a quintile long–short portfolio on the predicted surprise; it earns 0.63–0.73 percent per announcement against the equal-weighted benchmark and 1.35 percent against I/B/E/S (all  $p < 0.01$ ).<sup>9</sup>

A final test probes the GenAI channel directly. Following Cheng et al. (2025) and Liu and Subasi (2026), I use major ChatGPT outages as an exogenous shock to GenAI availability. On announcement days following a major outage, the accuracy premium against I/B/E/S collapses while the equal-weighted contrast shows no significant change. The asymmetry is what a GenAI-mediated integration channel predicts: when multi-FDP synthesis is unavailable, investors revert specifically to the I/B/E/S consensus.<sup>10</sup> An untabulated test of investor search behavior corroborates the route: after ChatGPT, Google search shifts toward the free forecast distributors GenAI cites rather than away from them, consistent with GenAI acting as a gateway to those distributors rather than a substitute for them.

This paper makes three contributions. First, I show that the market’s reliance on FDPs is not

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<sup>8</sup>This is not a foregone conclusion, given the within-provider null of Michaely et al. (2024) noted above.

<sup>9</sup>Specifically, the portfolio goes long the top quintile and short the bottom quintile of the standardized predicted surprise, holding each position over the four-day  $[-2, +1]$  announcement window. The per-announcement return is estimated by panel OLS with firm-clustered standard errors. Results are robust (untabulated) to alternative event windows, including  $[-2, +2]$  and  $[-1, +1]$ .

<sup>10</sup>Outages are sourced from OpenAI’s public status-page history; I require an incident of at least 240 minutes, yielding four incidents across five unique calendar days through 2024.

static but technology-dependent, extending recent evidence on cross-FDP differences in analyst consensus information (Larocque et al., 2025). When information-processing costs fall, the market reweights its implicit consensus toward the more accurate providers, with measurable effects on earnings-announcement returns. For researchers, the implication is methodological: the FDP whose consensus best reflects market expectations shifts with information technology.

Second, I add a distinct mechanism to the literature on the capital-market consequences of GenAI (e.g., Bertomeu et al., 2025; Blankespoor et al., 2026; Chang et al., 2026; Cheng et al., 2025; Liu and Subasi, 2026; Lopez-Lira and Tang, 2023; Xue et al., 2025). Prior work documents that GenAI lowers investor processing costs and shifts prices around specific events such as service outages or regulatory bans. The mechanism here is synthesis: GenAI makes it cheap for the market to combine forecasts across multiple FDPs, and this combined consensus is now priced into earnings-announcement reactions.

Third, I sharpen the disclosure-processing-cost framework of Blankespoor et al. (2020) by locating the cost component that prior single-source technologies left intact and that the market's accuracy-weighted use of consensus information required. A long line of work shows how prior single-source technologies reduce specific cost components: EDGAR for filing acquisition (Drake et al., 2015; Loughran and McDonald, 2017), XBRL for structured-data processing (Blankespoor, 2019), social media for awareness (Blankespoor et al., 2014), and robo-journalism for retail investors' processing costs (Blankespoor et al., 2018). The evidence here points to cross-source integration of heterogeneous consensus measures as that binding friction: the earlier technologies left it intact, and GenAI relaxes it, with measurable consequences for earnings-announcement pricing.

Section 2 reviews the related literature and presents the theoretical framework. Section 3 describes the institutional setting and data and constructs the accuracy-weighted surprise. Section 4 presents initial evidence that the market's reliance tracks provider accuracy and the consensus's predictive value after ChatGPT. Section 5 tests whether the post-ChatGPT market prices the accuracy-weighted synthesis across FDPs, documents its tradeable returns, traces the accuracy premium across GPT-capability regimes, and identifies its cross-sectional drivers. Section 6



shows that LLM visibility—a direct measure of GenAI forecast acquisition—is distinct from classical prominence proxies and gates the premium. Section 7 probes the GenAI channel directly by exploiting major ChatGPT outages as an exogenous shock to GenAI availability. Section 8 concludes.

## 2 Related Literature and Theoretical Framework

### 2.1 Forecast Data Providers and the Construction of Analyst Consensus

The market discriminates on forecast quality at the level of the individual analyst; no comparable discrimination has been documented at the level of the consensus. Sell-side analyst earnings forecasts have long served as the benchmark proxy for market expectations in research and practice (Bartov et al., 2002; Beyer et al., 2010; Bradshaw et al., 2017; Kothari, 2001), in part because analysts integrate firm-specific guidance, industry knowledge, and macroeconomic conditions in ways that simple time-series extrapolation does not, at least at short horizons (Bradshaw et al., 2012; Schipper, 1991). Individual forecast accuracy varies systematically with an analyst’s experience, employer resources, and portfolio complexity (Clement, 1999; Mikhail et al., 1997), and prices react more strongly to revisions from more accurate or higher-status analysts (Clement and Tse, 2003; Park and Stice, 2000; Stickel, 1992). At the consensus level, however, the analogous discrimination—building the consensus from the more accurate analysts—has not been shown to dominate the pooled benchmark, even within a single provider (Michaely et al., 2024). Understanding why discrimination stops at the analyst level begins with how the consensus is made.

The consensus number itself is not a primitive: third-party forecast data providers (FDPs) construct it by passing each underlying analyst forecast through proprietary screening and adjustment rules.<sup>11</sup> [Bochkay et al. \(2022\)](#) document the first-order capital-market consequences of these construction choices using Thomson Reuters’ 2009 methodology change for I/B/E/S.<sup>12</sup> An FDP’s

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<sup>11</sup>FDPs differ, among other dimensions, in (i) which broker estimates they include (in particular, when a forecast is deemed stale or noncomparable and dropped), (ii) how they treat nonrecurring items (some report GAAP-based earnings, others a “street” figure adjusted for one-time charges), and (iii) rounding conventions.

<sup>12</sup>Beginning in September 2009, Thomson Reuters stopped determining the I/B/E/S street-earnings actual from analysts’ ex post treatment of unexpected charges or gains reported in an earnings press release, adopting instead a uniform rule that immediately excludes such items from GAAP earnings. Because the change rolled out gradually, [Bochkay et al. \(2022\)](#) exploit the staggered variation in a difference-in-differences design.



treatment of unexpected charges and gains, they show, feeds directly into the time-series predictive properties of street earnings and into price discovery around announcements. The FDP layer, on this evidence, acts as a producer of information rather than a conduit for it.

The same construction discretion breeds pervasive disagreement across providers. Comparing the five FDPs in a large sample, [Larocque et al. \(2025\)](#) find that providers report different earnings surprises (beyond rounding) for the same firm-quarter 58 percent of the time, and that in 9 percent of firm-quarters at least one FDP signals a beat while at least one other signals a miss. These disagreements are associated with differences in price responsiveness, abnormal volatility, and liquidity around earnings announcements. The accuracy ranking across FDPs is itself unstable across firms and over time, so no fixed choice of provider reliably delivers the most accurate signal.

Disagreement of this magnitude leaves substantial room, in principle, for investors to synthesize across providers rather than anchor on a single one.<sup>13</sup> If the market could anticipate each FDP's accuracy ex ante, accuracy-weighted aggregation across providers could outperform any single-provider benchmark. Historically, the market appears not to have exploited this room. Although the most accurate provider varies by firm and period, [Larocque et al. \(2025\)](#) document that market reactions align most closely with the I/B/E/S surprise—the provider that ranks highest on their composite quality measures on average—consistent with I/B/E/S serving as a fixed anchor rather than a firm-by-firm choice.

The closest evidence on why the room went unexploited comes from [Michaely et al. \(2024\)](#), who examine the within-provider analog. A consensus built from high-quality analysts inside I/B/E/S yields no meaningful improvement in investor outcomes over the full-analyst consensus—a null they trace to three factors: error cancellation in the broader pool, imperfect persistence in analyst quality, and the cognitive cost of identifying high-quality forecasters.<sup>14</sup> Each of the three operates with greater force across providers, where accuracy rankings are unstable and methodologically

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<sup>13</sup>Classical forecast-combination theory ([Bates and Granger, 1969](#); [Clemen, 1989](#); [Timmermann, 2006](#)) establishes that the accuracy gain from combining forecasts rises as the forecasts' errors become less correlated; whether the gain is realizable in practice depends on the cost of locating and combining the underlying sources, a point developed below.

<sup>14</sup>This pattern echoes a broader theme in the analyst and disclosure literature, that how readily forecast information can be interpreted shapes how investors use it: forecast dispersion is itself a priced characteristic ([Diether et al., 2002](#)), and disclosure clarity is linked to investor behavior at the individual-disclosure level ([Lawrence, 2013](#)).

distinct consensus numbers must also be reconciled. The open question is whether a large enough reduction in the cost of identifying, acquiring, and integrating accurate sources can unlock the consensus-level discrimination that the historical environment failed to produce. The next subsection turns to the technologies that determine this cost.

## 2.2 Technological Advancements and Information Processing Costs in Capital Markets

Whether investors use available information depends on what it costs them to do so. [Blankespoor et al. \(2020\)](#) provide the canonical economic frame, decomposing the cost of using any piece of information into three components: *awareness* of its existence, *acquisition* of the information itself, and *integration* of that information into the investor’s decision. Each component is sensitive to technology, and an exogenous reduction in any one of them—holding the underlying information fixed—can measurably change investor behavior, with downstream consequences for asset prices and for firms’ own reporting choices.

An extensive empirical literature documents these effects component by component, with much of the identifying variation coming from single-source technological shocks introduced by regulators or by changes in market infrastructure. For awareness, [Blankespoor et al. \(2014\)](#) find that firms’ dissemination of news through Twitter narrows spreads and deepens markets, especially for less-visible firms—consistent with the news reaching a broader investor base. On the acquisition side, [Drake et al. \(2015\)](#) show that EDGAR search activity tracks firm-level information demand and that surges in retrieval precede price adjustments, while [Loughran and McDonald \(2017\)](#) use granular access logs to show that most filings draw few viewers but that the filings investors do open are associated with substantive price reactions. For integration, the leading example is the SEC’s XBRL mandate, which required firms to tag financial-statement items in machine-readable form: [Blankespoor \(2019\)](#) documents that the resulting reduction in investors’ processing costs shifted firms’ own disclosure choices. On the textual side, [Tetlock \(2007\)](#) shows that a dictionary-based measure of media pessimism extracts return-relevant information from a daily news column, [Loughran and McDonald \(2011\)](#) show that finance-specific word lists do the same for 10-K filings, and [Blankespoor et al. \(2018\)](#) demonstrate that robo-journalism (automated article generation)

lowers the processing cost retail investors face when consuming financial information.

A parallel literature approaches these costs from the demand side. [Da et al. \(2011\)](#) establish Google search as a measure of retail attention that predicts short-horizon returns, and [Drake et al. \(2012\)](#) document that Google search rises ahead of earnings announcements and that high pre-announcement search preempts part of the announcement-window price response. Attention is itself a binding constraint, with documented inattention along multiple margins: complicated firms ([Cohen and Lou, 2012](#)), competing simultaneous announcements ([Driskill et al., 2020](#); [Hirshleifer et al., 2009](#)), Friday timing ([DellaVigna and Pollet, 2009](#)), and strategic scheduling ([deHaan et al., 2015](#)). Scarce attention is what makes the recurring time cost of cross-FDP synthesis bind: the awareness, acquisition, and integration steps must be repeated for every firm-quarter an investor follows (Section 2.4), so an attention budget that is already stretched by a single firm’s announcement leaves little room for reconciling several providers’ numbers announcement by announcement.

Closest to the FDP-aggregation question, a disseminator-side literature ties broader access to firm disclosures to changes in price formation: conference calls open to all investors coincide with more small-investor trading and higher return volatility during the call ([Bushee et al., 2003](#)) and with reduced post-announcement underreaction ([Kimbrough, 2005](#)), and business-press coverage is associated with narrower spreads and deeper markets around earnings releases ([Bushee et al., 2010](#)).

A common limitation runs through this work: each advance lowered the cost of one component for a single information source. None enabled low-cost integration of numbers produced by several sources at once—combining, say, I/B/E/S, FactSet, Bloomberg, S&P Capital IQ, and Zacks consensus estimates in real time. That friction survived every one of these technologies.

## **2.3 Generative AI in Capital Markets**

GenAI is the first technology positioned to relax this cross-source friction. It arrives against the backdrop of a capital-markets machine-learning literature with one central finding: nonlinear, high-dimensional methods such as neural networks and tree-based models extract predictive signal that linear asset-pricing regressions miss ([Goldstein et al., 2021](#); [Gu et al., 2020](#)). What distinguishes the current generation of LLMs is the conjunction of three features prior technologies lacked:

they process unstructured information from heterogeneous sources, they synthesize it through natural-language queries without specialized retrieval infrastructure or programming expertise, and they operate at low marginal cost per query. Together, these features put cross-source synthesis—once the province of institutional teams with dedicated infrastructure—within reach of retail investors for the first time.

Early evidence indicates that this shift is already price-relevant. [Lopez-Lira and Tang \(2023\)](#) show that ChatGPT-derived sentiment signals predict cross-sectional returns; [Cheng et al. \(2025\)](#) use ChatGPT service outages as exogenous shocks and find that investor responses to information degrade when the tool is unavailable; [Bertomeu et al. \(2025\)](#) exploit Italy’s 2023 ChatGPT ban for country-level identification; and [Chang et al. \(2026\)](#) show that retail traders’ alignment with AI-derived earnings-call sentiment rose sharply after ChatGPT’s release, narrowing the information-processing gap between less-sophisticated retail traders and more-sophisticated short sellers.

Closer to the analyst forecasts studied here, a separate strand examines the production side—AI’s effects on analyst research itself. [Kimbrough et al. \(2026\)](#) find that analysts at brokerages with greater AI integration produce more accurate earnings forecasts and that their revisions are more informative to capital markets. AI-generated articles, [Bradshaw et al. \(2026\)](#) document, rose to 13.5 percent of Seeking Alpha’s output after ChatGPT’s release, with AI-using authors covering more firms but producing less informative content per article. On the crowdsourced side, [Liu and Subasi \(2026\)](#) find that GenAI closes the accuracy gap between nonprofessional and professional forecasters. Together, these papers indicate that AI is reshaping the production of analyst research. How GenAI reshapes the consumption of that research—in particular, how investors aggregate across the heterogeneous consensus numbers that competing FDPs construct—is the gap this paper fills.

## **2.4 Theoretical Framework and Predictions**

Two opposing forces shape the framework, and the balance between them determines whether consensus-level discrimination can emerge. The first comes from the forecast-combination literature:

combining heterogeneous, partially independent forecasts almost always dominates relying on any single source ([Bates and Granger, 1969](#); [Clemen, 1989](#); [Granger and Ramanathan, 1984](#); [Stock and Watson, 2004](#); [Timmermann, 2006](#)). The intuition is the wisdom-of-crowds principle popularized by [Surowiecki \(2004\)](#): averaging diversifies away idiosyncratic forecaster errors, so sub-selecting on individual accuracy can lose more from forgone diversification than it gains in precision. Applied to the analyst-consensus setting, this force rationalizes the historical equilibrium in which the market anchors on a single pooled benchmark (typically the I/B/E/S consensus)—itself already an average over many analysts—and treats cross-FDP differences as noise. It is also consistent with the within-source evidence: in [Ahlers and Lakonishok \(1983\)](#) and [Michaely et al. \(2024\)](#), sub-selecting forecasters on measured quality fails to deliver consistent gains over the broader pool. Taken alone, this force implies that the historical anchor should persist.

The opposing force is heterogeneity in what the providers produce: the documented cross-sectional and time-series variation in FDP accuracy ([Larocque et al., 2025](#)), together with the methodology-driven information content that [Bochkay et al. \(2022\)](#) identify. When providers disagree substantially, when part of that disagreement reflects accuracy rather than noise, and when the more accurate provider can be identified from its observable characteristics, a sufficiently low-cost synthesis across providers can dominate the pooled benchmark. The model-based earnings-forecast literature makes a parallel point: cross-sectional models deliver earnings forecasts with less bias and broader coverage than the analyst consensus, though not greater accuracy ([Hou et al., 2012](#)). The cost of identifying, acquiring, and integrating accurate sources is the state variable that determines which force the market can act on: when this cost is high, synthesis is infeasible whatever its statistical merit, and the market anchors on a pooled single-FDP consensus; when it is low, accuracy-weighted synthesis becomes feasible, and the ordering reverses if the accuracy-relevant heterogeneity across providers outweighs the diversification forgone by tilting away from the pool.

Historically, that cost was high: for cross-FDP synthesis, all three components of the [Blankespoor et al. \(2020\)](#) decomposition were costly at once, and integration—the component

every prior access technology left intact—was the binding one. Even when consensus data are freely redistributed through retail-facing platforms, an investor seeking the cross-FDP picture must know which platform carries which FDP’s consensus (awareness), visit each individually to retrieve the number (acquisition), and reconcile numbers that differ by methodology and are presented inconsistently across platforms (integration). These steps recur for every firm-quarter an investor follows, and synthesis at scale further demands programming and data-engineering expertise. Together, the recurring time cost and the expertise requirement confined the capability to specialists and anchored the market on a single source.

GenAI relaxes both barriers at once. A single LLM query identifies candidate FDP sources, retrieves their numbers, and integrates them into one response that reports the implied consensus and the cited distributors, all without specialized infrastructure or programming skill. The effective cost of cross-FDP synthesis collapses to that of issuing one query.<sup>15</sup> Figure 1 summarizes the framework, which yields two primary empirical predictions:

**Prediction 1.** After ChatGPT’s release, the consensus reflected in earnings-announcement prices shifts away from heuristic aggregation (single-FDP anchoring or equal-weighting across FDPs) toward a more sophisticated synthesis that draws on multiple FDPs and places greater weight on the more accurate ones.

**Prediction 2.** The post-ChatGPT shift is stronger when and where GenAI-mediated synthesis is more accessible (LLMs can retrieve the firm’s analyst forecasts) and more valuable (cross-FDP disagreement and its methodology-driven component are large).

The framework operates entirely on the demand side—a scope condition worth stating explicitly. Falling awareness, acquisition, and integration costs change how investors use the FDPs’ numbers, while the supply side—each provider’s construction rules and the resulting street earnings—is held fixed. The restriction separates the framework from a supply-side alternative in the spirit of the 2009 I/B/E/S methodology change studied by [Bochkay et al. \(2022\)](#): were the post-ChatGPT patterns

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<sup>15</sup>[Appendix B](#) reports two worked examples (IBM and Microsoft) showing the actual LLM responses, the cited distributors, and the upstream FDPs where disclosed.

instead driven by a change in how the street numbers are constructed, the predictive quality of street earnings itself—rather than the market’s response to it—would move at the ChatGPT boundary. A final prediction follows from reading Predictions 1 and 2 as statements about consensus *quality* rather than about trailing accuracy in particular. Accuracy is the dimension the weighting scheme uses, but a consensus can also be judged by its *predictive value*—how closely it anticipates the firm’s future street numbers. If the post-ChatGPT selectivity reflects a general taste for quality rather than an artifact of the trailing-accuracy construction, it should appear on this second dimension as well. Under the demand-side framework, what moves instead is the weight the market places on a consensus signal’s predictive value:

**Prediction 3.** The post-ChatGPT shift extends beyond trailing accuracy: announcement reactions load increasingly on consensus signals with greater predictive value for the firm’s future performance.

### 3 Institutional Background, Data, Sample, and Key Measures

The paper’s tests demand a specific empirical setting: providers that genuinely disagree, a panel that observes every provider’s consensus for the same firm-quarter, outcomes that register the market’s reaction, and a surprise benchmark that embeds provider accuracy. This section supplies each in turn.

#### 3.1 Institutional Background

Cross-FDP disagreement begins with industry structure. Five major forecast data providers—I/B/E/S, FactSet, Bloomberg, S&P Capital IQ, and Zacks—construct and sell analyst-consensus EPS information to sell-side brokerages, buy-side investors, financial media outlets, and academic researchers. I/B/E/S is credited with creating the consensus-earnings-estimate industry in the early 1970s, and Zacks claims to have first distributed quarterly consensus forecasts in the early 1980s. Bloomberg, Capital IQ, and FactSet entered the segment from adjacent product lines (terminal data, investment-banking software, and printed “fact sets” delivered to Wall Street



clients, respectively).<sup>16</sup>

Two sources of variation drive cross-FDP differences in consensus numbers. First, the underlying analyst set differs: each FDP includes a different subset of contributing sell-side analysts, applies its own recency cutoffs (e.g., Bloomberg drops estimates more than 30 days old), and runs proprietary statistical checks to filter outliers. Second, the methodology for constructing “street” earnings varies: each FDP decides for itself how to treat unexpected charges, gains, and other adjustments to reported earnings (Bochkay et al., 2022). Zacks, for example, counts stock-based compensation as a recurring expense in street earnings; I/B/E/S has historically excluded it. Such divergence is common: 46.7 percent of firm-quarters in the 2002–2020 sample of Larocque et al. (2025) contain a large unexpected item that FDPs treat inconsistently. Together, these coverage and methodology differences underlie the cross-FDP disagreement documented in Section 2.

Table 3 quantifies the methodology margin directly. For each pair of FDPs, it reports the share of firm-quarters in which the two providers publish *different* street actuals for the same reported quarter—a pure methodology comparison, since both providers observe the same GAAP filing and differ only in their construction rules. Even the closest pair, I/B/E/S and Capital IQ, publish different actuals in 14 percent of firm-quarters; Bloomberg’s actuals differ from every other provider’s in roughly 88 percent; and all five providers report the same actual in only 6.8 percent of firm-quarters, with the largest cross-provider gap reaching at least five cents per share in 77 percent. Disagreement at the actuals level implies disagreement in the surprise even for identical forecasts, so provider choice is consequential before any question of analyst composition arises.

No single FDP, however, strictly dominates the others at the firm-quarter level. I/B/E/S ranks highest on average across composite measures of FDP quality (Larocque et al., 2025) and has long served as the de facto anchor for research and trading alike, yet it is the most accurate provider in only 16.9 percent of firm-quarters in my S&P 1500 sample—below the 20 percent a uniform draw across five providers would imply—and the least accurate in 12.1 percent (untabulated). In the

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<sup>16</sup>Each FDP today operates within a broader financial-data platform: I/B/E/S under LSEG Data & Analytics (formerly Refinitiv); Capital IQ under S&P Global; FactSet, Bloomberg, and Zacks as independent firms. Bloomberg bundles earnings estimates with its core terminal offering, while the other four sell estimates as standalone or modular products. See Larocque et al. (2025) for a detailed institutional review.

remaining 71 percent of firm-quarters it is neither, and the extreme accuracy ranks are dispersed across the other four FDPs. This variation in accuracy leadership across firms and over time gives the cross-FDP synthesis question its economic content: a static anchoring choice cannot exploit it; an adaptive multi-FDP rule can.

### 3.2 Data and Sample

Exploiting this firm-quarter variation requires parallel consensus histories from all five FDPs, which no single database supplies. I therefore collect quarterly consensus EPS forecasts and reported actuals from each provider separately. I/B/E/S and Zacks are available through WRDS (the I/B/E/S Summary History file and the Zacks Estimates database, respectively). The remaining three FDPs (FactSet, Bloomberg, and S&P Capital IQ) sit behind proprietary terminals or institutional portals, from which I download quarterly consensus histories firm by firm.<sup>17</sup>

The initial sample comprises all S&P 1500 firm-quarters with an earnings announcement over 2021–2025: 29,395 firm-quarters across 1,502 firms.<sup>18</sup> I then require complete coverage across all five FDPs, which removes 4,368 firm-quarters lacking a consensus from at least one provider; this screen is applied to the announcement quarter’s own consensus values. The accuracy weights of Section 3.4 additionally draw on each provider’s forecast errors over the four quarters preceding the announcement, so the consensus and actual histories underlying the weights reach back before the 2021 start of the announcement window, into 2020, for the earliest sample announcements; the resulting accuracy-weighted composite is non-missing for every firm-quarter in the final sample (Table 2). Dropping firm-quarters with missing market-outcome variables or controls (1,179 observations), missing constructed return or predicted-surprise variables (10), and singleton firms (5) brings the final sample to 23,833 firm-quarters across 1,334 firms. Table 1 details the full selection procedure; Table 2 reports descriptive statistics for the main variables.<sup>19</sup>

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<sup>17</sup>Larocque et al. (2025) follow a similar collection procedure for their 2002–2020 study, downloading or hand-collecting data from each FDP’s proprietary system. Bloomberg consensus data, in particular, must be retrieved ticker by ticker through Bloomberg Query Language (BQL) functions, and its historical consensus reaches back only five years, which constrains the sample start date.

<sup>18</sup>The November 30, 2022 ChatGPT release falls within this window, leaving roughly two years of pre-ChatGPT and three years of post-ChatGPT announcements—a relatively balanced panel of firm-quarters. The five-year window is also the maximum permitted by Bloomberg’s historical-retrieval constraint described above.

<sup>19</sup>All continuous variables in the analyses that follow are winsorized at the 1st and 99th percentiles.

Firm controls and market-outcome variables are drawn from WRDS-hosted databases: accounting fundamentals from Compustat, returns and trading data from CRSP, institutional ownership from the Thomson Reuters 13F Holdings database, and analyst-coverage characteristics from I/B/E/S. The control vector comprises log market equity, book-to-market, leverage, the prior twelve-month return, log price, return volatility, analyst dispersion, announcement-day busyness, the absolute changes in income before extraordinary items and in shares outstanding, the magnitude of special items, an indicator for high stock-based compensation, indicators for unexpected items, stock splits, and fiscal fourth quarters, institutional ownership, and the logged number of analysts (Appendix A, Panel B). Appendix A defines all variables.

### 3.3 Dependent Variables

The market's reliance on any given consensus is not directly observable, so I infer it from the price response at the earnings announcement: the surprise implied by a consensus on which the market relies more heavily should move prices more. The main outcome variables are short-window abnormal returns: a three-day cumulative abnormal return  $CAR_{[-1,+1]}$  for the immediate-reaction tests and a four-day buy-and-hold abnormal return  $BHAR_{[-2,+1]}$  for the trading-strategy test.<sup>20</sup> For each window I use two risk adjustments: a Fama-French-Carhart four-factor (FFC) abnormal return and a market-adjusted return.<sup>21</sup>

### 3.4 Constructing the Accuracy-Weighted SUE

The paper's central construct is an accuracy-weighted aggregation of the five FDP consensus signals. Let  $g \in \{\text{I/B/E/S, FactSet, Bloomberg, Capital IQ, Zacks}\}$  index providers and let  $SUE_{i,q}^g = (actual_{i,q}^g - consensus_{i,q}^g) / p_{i,q}$  denote provider  $g$ 's standardized unexpected earnings for firm  $i$  in quarter  $q$ : its own street actual less its own consensus, scaled by share price  $p_{i,q}$ . The

<sup>20</sup>Relative to the three-day CAR window, the four-day BHAR window opens one day earlier, at day -2, so that a position taken ahead of the announcement captures any pre-announcement leakage; both windows close at day +1, which captures the next session's reaction to announcements made after the market close.

<sup>21</sup>FFC factor loadings are estimated on the [-260, -11] pre-announcement window; the market-adjusted return subtracts the contemporaneous CRSP value-weighted index return from each daily return. All returns are expressed in percent. Inferences are unchanged under either adjustment: Tables 5 and 7 report both side by side, as do Table 4, Panel B, and Table 6, Panel B, while Tables 4 (Panel A), 8, 9, 10, 11, and 12 report only the market-adjusted return for brevity (the FFC counterparts are available on request).

accuracy-weighted composite is

$$SUE_{i,q}^{Acc} = \sum_g w_{i,q}^g SUE_{i,q}^g, \quad (1)$$

with weights  $w_{i,q}^g$  derived from each provider's recent forecast accuracy. The equal-weighted benchmark sets  $w_{i,q}^g = 1/5$ , yielding  $SUE_{i,q}^{EW5}$ ; the single-FDP I/B/E/S benchmark,  $SUE_{i,q}^{IBES}$ , requires no aggregation. Entering the accuracy-weighted composite alongside either benchmark in the same regression tests whether the market loads more on the accuracy-tilted aggregate than on the heuristic alternative.

*Weight construction.* The weights map each provider's recent, firm-specific track record into its share of the composite. I first compute the price-scaled absolute forecast error,  $err_{i,q}^g = |actual_{i,q}^g - consensus_{i,q}^g|/p_{i,q}$ , where  $consensus_{i,q}^g$  is provider  $g$ 's end-of-quarter consensus for quarter  $q$ , observed before the announcement, and  $p_{i,q}$  is the pre-announcement share price that also scales the surprise itself,  $SUE_{i,q}^g = (actual_{i,q}^g - consensus_{i,q}^g)/p_{i,q}$ , expressed in percent; the error and the surprise are built from the same consensus and price.<sup>22</sup> A backward-looking four-quarter rolling mean then yields a recent mean absolute forecast error,  $MAFE_{i,q}^g = \frac{1}{4} \sum_{k=1}^4 err_{i,q-k}^g$ , whose sign I flip so that larger values denote greater accuracy:  $acc_{i,q}^g = -MAFE_{i,q}^g$ . Because the rolling window excludes quarter  $q$  itself, the weights use only information available before the quarter- $q$  announcement and cannot induce look-ahead bias. I standardize across the five providers within firm-quarter,  $acc\_z_{i,q}^g = (acc_{i,q}^g - \overline{acc}_{i,q})/sd(acc_{i,q})$ , and convert the  $z$ -scores into convex weights via the softmax (multinomial-logit) transform,

$$w_{i,q}^g = \frac{\exp(acc\_z_{i,q}^g)}{\sum_h \exp(acc\_z_{i,q}^h)}. \quad (2)$$

The resulting weights are strictly positive, sum to one within each firm-quarter, preserve the accuracy ordering, and reduce to the uniform 0.2 vector when all five FDPs are equally accurate. When the five accuracies coincide, the within-row standard deviation is zero and the weights are set

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<sup>22</sup>The reported actual EPS is itself FDP-specific: each provider computes its own street earnings actual under its own screening and adjustment rules. Pairing each FDP's consensus with its own actual ensures that the resulting forecast error reflects only ex ante predictive accuracy, not methodology mismatch between the consensus and the realized value. This is the approach taken by [Larocque et al. \(2025\)](#) and is consistent with how each FDP reports its own pre-announcement consensus relative to its own post-announcement actual.

to the uniform vector by convention. [Appendix C](#) works through this construction for a single hypothetical firm-quarter, tracing how raw accuracy histories pass through each step—error, rolling MAFE, within-firm-quarter standardization, and softmax—to produce both the weights  $w_{i,q}^g$  and the composite  $SUE_{i,q}^{Acc}$ .

*Design choices.* Three choices shape the weights. First, the within-firm-quarter standardization (rather than standardization pooled across all firm-quarters) is essential because earnings noise differs across firms and across periods: what matters economically is relative accuracy for this firm this quarter, not absolute accuracy compared with other firms or other quarters. Second, the softmax transform, akin to the exponential weighting schemes in the forecast-combination literature (e.g., [Bates and Granger, 1969](#); [Timmermann, 2006](#)), occupies a middle ground between equal weighting (which ignores accuracy) and winner-take-all selection (which would amplify the noise in the four-quarter MAFE estimate). It also admits a maximum-entropy interpretation: among weight vectors delivering the same weighted-average standardized accuracy, the softmax vector imposes the least additional structure. Third, the backward-looking four-quarter window balances recency against statistical noise: shorter windows overweight transient accuracy shocks; longer windows blur structural changes in provider quality.

*Empirical weight distribution.* In practice, the weights tilt meaningfully away from equal weighting while stopping well short of winner-take-all. The average within-firm-quarter minimum weight is 0.028 and the average maximum is 0.414 (untabulated), against a uniform benchmark of 0.200: the typical firm-quarter cuts its least accurate provider to roughly 3 percent of the composite and concentrates roughly 41 percent on its most accurate.  $SUE^{Acc}$  and  $SUE^{EW5}$  are highly correlated ( $\rho \approx 0.82$ , untabulated) but not perfectly collinear—leaving distinct variation for the pricing tests to exploit.

#### 4 GenAI and Selective Reliance Across FDPs: Initial Evidence

This section presents the first evidence on [Prediction 1](#): if GenAI lets the market discriminate across FDPs, the price reaction to a given surprise should rise with the accuracy of the provider that produced it, and rise more steeply after ChatGPT. Three tests implement this logic: a

within-firm-quarter test of whether the earnings response rises with provider accuracy rank; a firm-level test of whether reliance on I/B/E/S, the historical anchor, moves with I/B/E/S's own record; and a test of whether the same selectivity appears on a second quality dimension, the consensus's predictive value ([Prediction 3](#)).

*Design.* The first test asks whether the ERC rises with provider accuracy rank. Ranking the five FDPs within each firm-quarter by ex ante trailing MAFE (rank 1 = least, 5 = most accurate) and stacking the five provider observations for each firm-quarter, I estimate

$$CAR_{i,q,[-1,+1]}^{Mkt} = \alpha + \beta SUE_{i,q}^g + \delta Rank_{i,q}^g + \gamma (SUE_{i,q}^g \times Rank_{i,q}^g) + \varepsilon_{i,q,g}, \quad (3)$$

where  $\gamma$  is the ERC increment per one-rank accuracy gain.<sup>23</sup> Because rank is assigned within firm-quarter, identification comes from the same announcement return's differential sensitivity to the five providers' surprises: firm- and time-level confounders, including methodology drift common to all providers, difference out. The design cannot show, however, whether the market's reliance on a given provider rises and falls with that provider's record.

The second test asks exactly that: it fixes the provider and exploits cross-firm variation in how its accuracy rank changed around ChatGPT. Classifying firms by the change in I/B/E/S's mean rank between the pre- and post-ChatGPT windows ( $IBESUp_i$  and  $IBESDown_i$  mark improvements and deteriorations of  $\geq 2$  positions), I estimate the triple-interaction specification

$$\begin{aligned} CAR_{i,q} = & \alpha + \beta_1 SUE_{i,q}^{IBES} + \beta_2 SUE_{i,q}^{IBES} \times Post_q \times IBESUp_i \\ & + \beta_3 SUE_{i,q}^{IBES} \times Post_q \times IBESDown_i + \text{controls} + \text{FE} + \varepsilon_{i,q}, \end{aligned} \quad (4)$$

with the lower-order terms included but not displayed. The triple interactions carry the test. Selective reliance predicts  $\beta_3 < 0$ ; and if the pre-ChatGPT market already overweighted its anchor, a gain in I/B/E/S's rank adds little, so  $\beta_2 \approx 0$ .

*Results.* Post-ChatGPT, the price reaction to a given surprise depends increasingly on which provider produced it. [Table 4](#) reports both tests. In Panel A, the per-rank ERC increment  $\gamma$

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<sup>23</sup>The earnings response coefficient (ERC) is the slope of announcement returns on the earnings surprise—the abnormal return the market delivers per unit of surprise. A larger ERC means the market treats the surprise as more informative;  $\gamma$  measures how much that slope rises when the surprise comes from a one-rank-more-accurate provider.

roughly doubles, from 0.306 before to 0.621 after (both  $p < 0.01$ ); the post-pre differential is 0.316 ( $p < 0.01$ ). The provider rows show the breadth of the pattern: the post-ChatGPT per-rank increment is positive and significant for I/B/E/S (0.295), FactSet (0.380), Bloomberg (0.162), and Capital IQ (0.349, already 0.543 before ChatGPT), but not for Zacks (-0.020).<sup>24</sup> Panel B shows that the adjustment tracks I/B/E/S’s own record: the response to the I/B/E/S surprise weakens significantly where I/B/E/S *lost* rank—the post-ChatGPT triple interaction for I/B/E/S-down firms ranges from -2.3 to -2.8 across specifications ( $p < 0.05$ )—but not where it gained or held.<sup>25</sup> The reweighting is thus asymmetric, unwinding reliance where the historical anchor’s accuracy advantage has eroded while adding none where it grew—consistent with a pre-ChatGPT market that already overweighted I/B/E/S.

*A second quality dimension: predictive value.* Trailing accuracy is one notion of consensus quality; a complementary notion is *predictive value*—how closely the consensus anticipates the firm’s future street numbers. If the post-ChatGPT selectivity documented above reflects a general taste for consensus quality rather than a peculiarity of the trailing-accuracy construction, the same pattern should appear on this dimension as well ([Prediction 3](#)).

For each firm, I score the consensus’s predictive record in real time. The building block is the own-basis anticipation error: the absolute per-share distance between the quarter- $q$  I/B/E/S consensus and the seasonally matched quarter- $(q+4)$  street actual, excluding or adjusting pairs that straddle a stock split—“own basis” because the consensus is paired with the same provider’s street actual. I average this error over up to the twenty most recent source quarters whose year-ahead actuals are already public at the announcement (at least twelve required), scale it by the window mean  $|actual|$  so it is comparable across firms, and winsorize. The measure is then  $Predictive\ Value_{i,q}^{IBES} = 1 - error_{i,q}$ , a continuous cross-firm variable (mean 0.56, standard deviation 0.30): higher values mean the firm’s consensus has anticipated its year-ahead street numbers more closely. It covers 22,364 of the 23,833 firm-quarters (8,730 pre-ChatGPT), of which 22,356 enter the regressions;

<sup>24</sup>Versions of Panel A estimated cell by cell, one regression per provider–rank pair, in both absolute-value and signed form, yield the same inferences (untabulated).

<sup>25</sup>A generalized post-ChatGPT shift in risk premia or attention would move all providers’ ERCs together; neither the within-announcement rank gradient nor the cross-firm asymmetry fits that account.



the uncovered firm-quarters are recent listings with fewer than twelve scorable history quarters. If investors now read earnings through tools that weight the consensus by its demonstrated predictive value, the ERC should rise more steeply with *Predictive Value*<sup>IBES</sup> after ChatGPT.

The market prices the predictive-value dimension, and does so more steeply after ChatGPT. In Table 5, the ERC gradient on predictive value is already strongly positive before ChatGPT (2.17,  $t = 4.0$ )—firms whose consensus predicts well carry higher ERCs, a familiar association (Collins and Kothari, 1989)—so the identified claim is the post-ChatGPT increment: +2.05 ( $t = 2.68$ ) market-adjusted and +2.10 ( $t = 2.82$ ) under the FFC adjustment. The increment matches the pre-period gradient in magnitude, so the slope roughly doubles after ChatGPT. Because the measure is close to a firm’s earnings predictability, the increment also admits a reading without GenAI—the persistence–ERC relation could have steepened at the same boundary for other reasons, such as the 2022 shift in discount rates—a possibility the controls below do not address directly. Dispersion, volatility, and the unexpected-item flag—the characteristics most likely to stand in for predictability—are among the controls, and the *Predictive Value*<sup>IBES</sup> level terms are indistinguishable from zero: well-predicted firms earn no different announcement return per se, so the effect runs entirely through the surprise loading. Column (3) splits the increment across the three post-ChatGPT GPT-capability regimes (text-only from November 30, 2022; browsing from September 27, 2023; native search from October 31, 2024; Section 5): 0.92 ( $t = 0.9$ ) under text-only, 1.44 ( $t = 1.5$ ) under browsing, and 3.54 ( $t = 3.1$ ) under native search. The point estimates rise with the tools’ retrieval capabilities, though only the native-search step is significant at conventional levels (in column (4), the browsing increment turns marginally significant under the Fama-French-Carhart adjustment)—the profile expected for a forecast track record that only browsing and native search made retrievable. Because the native-search regime spans only the final quarters of the sample, the regime profile is descriptive; the pooled increment in columns (1)–(2) is the formal test.

Two alternative readings of the increment do not survive. Announcement composition does not account for it: adding the large-unexpected-item flag of [Bochkay et al. \(2022\)](#) with its full set of interactions leaves the increment unchanged (2.04,  $t = 2.67$ ), and the flag is essentially



uncorrelated with predictive value (-0.02). Nor does selection toward I/B/E/S specifically: in an untabulated decomposition of the surprise into the component common to all providers and the I/B/E/S-specific tilt, the common component carries both the gradient (2.3,  $t = 3.7$ ) and its post-ChatGPT increment (1.9,  $t = 2.0$ ), though the increment clears the conventional threshold only narrowly. The post-ChatGPT market loads more heavily on the consensus signal of well-predicted firms, whichever provider supplies it.

The provider level points the same way, but the evidence there is fragile. A within-firm-quarter rank version of the score across four providers (each provider’s own-basis error against its own year-ahead actual; Bloomberg is excluded because its native point-in-time history begins only in 2020Q2) shows no pre-period rank gradient (untabulated), a post-period gradient of 0.320 ( $t = 4.14$ ), and a shift of 0.242 ( $t = 2.56$ ). But the shift’s significance concentrates in the Capital IQ observations, and the predictive-value and accuracy rankings are statistically indistinguishable once corrected for the low reliability of each, so accuracy weighting already embeds much of the provider-level predictive-value signal.<sup>26</sup>

Beyond extending the evidence, the predictive-value measure settles which quality dimension the weighting analyses should use. Applied at the current-quarter horizon—the quarter- $q$  I/B/E/S consensus against the quarter- $q$  street actual, with the same twenty-quarter window and  $|actual|$  scaling—the same procedure yields a like-for-like measure of the consensus’s trailing *accuracy* that differs from predictive value only in the horizon of the actual it targets. Across firms the two are highly correlated (Pearson 0.72, Spearman 0.78; 0.82 correcting for split-half measurement error; stable pre- and post-ChatGPT and within announcement quarters). Predictive value and accuracy are thus two horizons of one underlying quality—how well the firm’s consensus predicts its street numbers. This like-for-like measure is not the accuracy input of the weighting analyses, which is the four-quarter, price-scaled  $MAFE_{i,q}^s$  of Section 3.4; the correlations above therefore speak to the

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<sup>26</sup>Two caveats temper the reading. First, the own-basis pairing rewards internal consistency of street-earnings rules alongside genuine foresight about the firm, and the level gradient partly reflects earnings predictability—hence the identified claim is confined to the post-ChatGPT increment. Second, the rank version’s split-half reliability is below 0.35; excluding Capital IQ, its shift is 0.099 ( $t = 0.75$ ), while excluding any other provider leaves it at 0.26–0.31 (untabulated).

overlap between the two quality dimensions and do not by themselves measure the overlap between predictive value and the four-quarter MAFE the weights use. The weighting analyses that follow operate on accuracy for two reasons: the current-quarter error is realized at every announcement rather than after a four-quarter maturation, so accuracy is observable in real time; and it varies firm-quarter by firm-quarter, supplying the variation the weighting scheme exploits—while, on the like-for-like evidence above, the accuracy dimension carries most of the predictive-value content. The next section asks whether the market goes further and prices an accuracy-weighted synthesis across providers.

## 5 GenAI and the Accuracy-Weighted Synthesis Across FDPs

This section tests the paper’s central claim, stated in [Prediction 1](#): after ChatGPT, the market prices the accuracy-weighted synthesis of all five FDPs beyond what the heuristic alternatives capture. Section 4 establishes the precondition—the post-ChatGPT market discriminates across FDPs by accuracy—but discrimination is a weaker claim than synthesis, because a market can favor a single accurate provider without ever combining across providers. The analysis proceeds in three steps. I first validate the construct itself: accuracy ranks persist, the accuracy-weighted composite is the most accurate of the three, and its *ex ante* difference from the heuristics earns an *ex post* trading return. A horse race then pits the accuracy-weighted SUE against the two heuristic benchmarks; the resulting accuracy premium is statistically and economically significant after ChatGPT. Finally, testing [Prediction 2](#), I trace the premium across the four GPT-capability regimes and locate its cross-sectional concentration where the framework predicts: in firm-quarters with greater cross-FDP disagreement and greater methodological divergence across providers.

### 5.1 Validating the Accuracy-Weighted Construct

*Premise and benefits.* The design rests on one premise: weighting by *trailing* accuracy is informative only if accuracy *persists*. Given persistence, two benefits should follow:  $SUE^{Acc}$  should be more accurate than the heuristics, and it should carry trading value—its difference from a

heuristic, known ex ante, earning an ex post return. For the latter I form the *predicted surprise*

$$Pred.Surp_{i,q}^X = (Consensus_{i,q}^{Acc} - Consensus_{i,q}^X) / p_{i,q}, \quad X \in \{EW5, IBES\}, \quad (5)$$

where  $Consensus_{i,q}^{Acc} = \sum_g w_{i,q}^g consensus_{i,q}^g$  is the consensus EPS built with the weights of Equation (2),  $Consensus_{i,q}^X$  is the corresponding heuristic consensus, and  $p_{i,q}$  is the pre-announcement share price. I min-max standardize the predicted surprise to the unit interval and test whether it forecasts the four-day announcement BHAR and whether a long–short portfolio sorted on its quintiles earns a positive return.

*Results.* The premise holds, and so do both benefits. Provider ranks are highly persistent (Spearman  $\rho = 0.71$ ): in Table 6, Panel A, the least- and most-accurate providers retain their ranks 76 and 65 percent of the time, roughly three to four times the uniform baseline. Part of this persistence is mechanical, because adjacent four-quarter windows share three quarters, so the sharper validation is whether the trailing rank forecasts accuracy out of sample. In Figure 2,  $SUE^{Acc}$  has the smallest mean absolute error of the three surprise measures in both the pre- and post-ChatGPT periods. Panel (a) also shows surprise magnitudes declining over the sample for all three measures; because ERCs are concave in surprise magnitude, part of the post-ChatGPT rise in every loading may reflect smaller surprises, which is why the tests below rest on the contrast between composites estimated on the same announcements rather than on the level of any loading. In untabulated paired tests, the accuracy-weighted composite’s absolute standardized surprise is smaller than either heuristic’s in both periods ( $t > 10$ ). And in an untabulated stacked provider panel that regresses four-quarter-ahead earnings and cash flows on each provider’s consensus (the four providers with native point-in-time history, since Bloomberg’s begins in 2020Q2; provider-specific intercepts, industry effects, and firm-clustered standard errors), the consensus slopes are statistically indistinguishable across providers, so tilting weights toward accurate providers forgoes no predictive content. Trading value materializes in Panel B: the standardized predicted surprise forecasts the four-day BHAR in all four columns ( $p < 0.01$ ); per standard deviation of the regressor (0.116 and 0.109 in Table 2), the market-adjusted slopes imply BHAR increments of about 0.29 and 0.37 percentage points against the equal-weighted and I/B/E/S benchmarks, and a long–short portfolio on

its quintiles earns 0.63–0.73 percent per announcement against equal-weighting and 1.35 percent against I/B/E/S ( $p < 0.01$ ).

## 5.2 The Accuracy Premium: Horse-Race Tests

*Design.* With the construct validated, the horse race asks whether the market prices it. I race the accuracy-weighted composite  $SUE^{Acc}$  against two natural benchmarks: the equal-weighted composite  $SUE^{EW5}$ , which treats each FDP as equally informative, and the I/B/E/S-only consensus  $SUE^{IBES}$ , the historical anchor. The pooled specification is

$$\begin{aligned} CAR_{i,q} = & \alpha + \beta_1 SUE_{i,q}^{Acc} + \beta_2 SUE_{i,q}^X + \beta_3 SUE_{i,q}^{Acc} \times Post_q \\ & + \beta_4 SUE_{i,q}^X \times Post_q + \text{controls} + \text{FE} + \varepsilon_{i,q}, \end{aligned} \quad (6)$$

for  $X \in \{EW5, IBES\}$ . The pre-period accuracy premium is the contrast  $\beta_1 - \beta_2$ ; its post-pre change,  $\beta_3 - \beta_4$ , is the central test of [Prediction 1](#). A positive estimate of  $\beta_3 - \beta_4$  indicates that after ChatGPT the market’s loading shifts toward the accuracy-weighted consensus relative to the heuristic benchmark. The dependent variable is the three-day CAR under both risk adjustments, market-adjusted and FFC four-factor. The premium  $\beta_1 - \beta_2$  and its change  $\beta_3 - \beta_4$  are the estimands for every test in this section: the regime and cross-sectional analyses re-estimate them within subperiods and subsamples.<sup>27</sup>

*Results.* After ChatGPT, the market tilts toward the accuracy-weighted composite at the expense of both heuristics (Table 7). In Panel A (equal-weighted benchmark), the pre-ChatGPT accuracy premium is statistically indistinguishable from zero (-0.948), consistent with the equal-weighted composite already capturing the pre-period information set. The premium then turns sharply positive: the accuracy-weighted slope rises from 1.376 to 3.420 while the equal-weighted slope falls from 2.324 to 1.767. The formal test is the pooled interaction contrast, and the premium’s

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<sup>27</sup>Although  $SUE^{Acc}$  and the benchmark  $SUE^X$  are weighted averages of the same five per-FDP surprises and are therefore correlated by construction ( $\rho \approx 0.82$  for  $X = EW5$ ), they are not collinear: with  $R_j^2 \approx 0.67$ , the variance inflation factor  $VIF_j = 1/(1-R_j^2) \approx 3.0$ , so each individual slope’s standard error is only  $\sqrt{3.0} \approx 1.7$  times its single-regressor counterpart. The identifying quantity, however, is the contrast  $\beta_1 - \beta_2$ , and because positively correlated regressors yield negatively correlated slope estimates, its variance  $\text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\beta}_2) - 2\text{Cov}(\hat{\beta}_1, \hat{\beta}_2)$  exceeds the sum of the two individual variances. That larger variance is a property of the estimand, not of the horse-race parameterization: any invertible linear reparameterization of  $SUE^{Acc}$  and  $SUE^X$  (and of their  $Post_q$  interactions) fits the same model and returns the contrast with the same standard error. Entering the accuracy tilt  $SUE^{Acc} - SUE^X$  alongside the average  $(SUE^{Acc} + SUE^X)/2$ , for instance, returns the contrast as twice the tilt’s slope with an identical  $t$ -statistic, whereas entering the tilt alongside the benchmark  $SUE^X$  returns  $\beta_1$ , not the contrast, as the tilt’s slope. Because every test in this section rests on the contrast, the inflated standard errors of the individual slopes do not bear on the inference.

post-pre change is 2.401 ( $p < 0.01$ ) under the market-adjusted CAR and 2.584 ( $p < 0.01$ ) under the Fama-French-Carhart four-factor abnormal return. Panel B (I/B/E/S benchmark) adds the complementary result. In the pooled specification, the pre-ChatGPT accuracy premium is negative (-0.747 in the pre-period subsample, column (1); -0.623 in the pooled specification, column (3)) but statistically insignificant—a sign consistent with the pre-ChatGPT overweighting of I/B/E/S inferred in Section 4—and the post-ChatGPT premium climbs to 2.049 ( $p < 0.01$ ), a post-pre change of 2.672 ( $p < 0.01$ ) under the market-adjusted CAR and 2.688 ( $p < 0.01$ ) under the Fama-French-Carhart adjustment.

The magnitudes are economically large. After ChatGPT, the response to a one-percentage-point standardized surprise rises by roughly 2.4 percentage points more under the accuracy-weighted consensus than under the equal-weighted benchmark, and by 2.7 more than under I/B/E/S—increments comparable to the entire pre-ChatGPT ERC on the I/B/E/S consensus itself (2. Scaled by a typical tilt, the magnitudes are more modest: with a post-ChatGPT premium of 1.43 against the equal-weighted benchmark and a tilt standard deviation of roughly 0.29 percent of price (implied by the Table 2 standard deviations and the 0.82 correlation), a one-standard-deviation tilt moves the three-day CAR by about 0.4 percentage points, roughly five percent of the CAR’s standard deviation.<sup>2</sup> in Panel B). Put differently, the extra weight the post-ChatGPT market places on the accuracy-weighted synthesis over I/B/E/S is about the size of the entire weight it once placed on I/B/E/S. The shift tracks real informational content: the measure the market now favors is also the most accurate of the three (Figure 2). The same figure closes off a mechanical reading of the widening: a more accurate composite proxies better for the market’s expectation and would earn a higher slope for that reason alone, but its accuracy advantage over each heuristic is essentially unchanged from the pre- to the post-ChatGPT period (Panel (b)), so a better proxy cannot explain a premium that widens.<sup>28</sup> Yet the pooled contrast dates the shift only to the ChatGPT boundary; attribution to GenAI demands a finer clock.

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<sup>28</sup>The accuracy-weighted construction is a tractable proxy for the broader hypothesis that the post-ChatGPT market weights providers by accuracy, not a claim that the market literally applies softmax weights. Formally, the accuracy tilt  $SUE_{i,q}^{Acc} - SUE_{i,q}^{EW5} = \sum_g (w_{i,q}^g - 1/5) SUE_{i,q}^g$  is the inner product of each provider’s deviation from the uniform weight and its standardized surprise; this is the object the horse-race contrast identifies.

### 5.3 The Accuracy Premium Across GPT-Capability Regimes

*Motivation.* [Prediction 2](#) supplies that clock: if the premium reflects GenAI-mediated synthesis, it should strengthen as the tools’ retrieval capabilities expand—a dose–response pattern rather than a single jump at the ChatGPT release. I therefore re-estimate the horse race of Equation (6) within four GPT-capability regimes: pre-ChatGPT (R0), text-only (R1, from November 30, 2022), browsing (R2, from September 27, 2023), and native search (R3, from October 31, 2024).

*Results.* Table 8 and Figure 3 sharpen the timing. Against I/B/E/S, the accuracy premium’s point estimates are negative in the pre-ChatGPT (-0.75) and text-only (-1.23) regimes and turn positive under browsing (1.52); only under native search is the estimate statistically significant (4.91,  $p < 0.01$ ). Against the equal-weighted benchmark, the path is -0.95, 0.18, 2.43 ( $p < 0.05$ ), and 2.55 ( $p < 0.05$ ), so significance arrives at the browsing step and persists thereafter. Once retrieval arrives, the point estimates climb the staircase that [Prediction 2](#) anticipates, and the text-only regime supplies the telling null: a chatbot that cannot yet retrieve forecasts leaves the premium statistically undetectable against either benchmark. The predictive-value gradient of Section 4 climbs the same steps (Table 5, columns (3)–(4)), with only its native-search increment significant. In both series, statistical detection waits for retrieval.

### 5.4 Cross-Sectional Tests of the Accuracy Premium

*Design.* [Prediction 2](#) specifies where as well as when: the premium should concentrate in firm-quarters where GenAI-mediated synthesis is more valuable. Because the regime evidence localizes the accuracy premium to the browsing and native-search regimes, the cross-sectional tests replace the post-ChatGPT indicator  $Post_q$  with the post-scraping indicator  $Post_q^{Scr}$ —equal to one for announcements on or after the onset of GPT browsing (September 27, 2023), when web retrieval first made cross-provider synthesis feasible. I re-estimate the horse-race specification in Equation (6) within the below- and above-median halves of two partitions, each probing a distinct channel. (i) *Concurrent cross-FDP SUE dispersion* is the interquartile range of the five FDPs’ SUE for the same firm-quarter—the setting in which provider weighting has the most discriminating work to do. (ii) *Special-items magnitude* is the absolute value of special items scaled by the absolute

value of income before extraordinary items; because FDPs differ in their treatment of unexpected charges and gains, cross-FDP disagreement over what belongs in street earnings should be largest where such items are economically material.<sup>29</sup>

*Results.* Table 9 reports the partition-by-partition estimates; the post-scraping change in the premium is significant only in the predicted half of each partition. In Panel A (cross-FDP dispersion), the post-pre change in the accuracy premium is 2.211 ( $p < 0.05$ ) against the equal-weighted benchmark and 2.427 ( $p < 0.01$ ) against I/B/E/S in the high-disagreement subsample, versus insignificant estimates (2.037 and -0.691) in the low-disagreement subsample. A caveat attends the equal-weighted comparison: the low-disagreement point estimate is close in magnitude to its high-disagreement counterpart and differs mainly in precision; the partition separates cleanly only against I/B/E/S, where the low-disagreement estimate turns negative (in that subsample  $SUE^{Acc}$  and  $SUE^{IBES}$  nearly coincide, so the two loadings are separately ill-determined and the level of the premium is a collinearity artefact; its post-pre change, -0.691, is insignificant). Panel B (special items) requires no such qualification: the premium change is 2.518 ( $p < 0.05$ ) against the equal-weighted benchmark and 3.638 ( $p < 0.01$ ) against I/B/E/S in the high-special-items subsample, versus point estimates of the opposite sign (-0.552 and -0.219) in the low-special-items subsample. The two partitions speak to different parts of the mechanism. The concentration in high-dispersion firm-quarters supports the discrimination claim: the premium is statistically detectable only where the five FDPs disagree most about the size of the surprise. The concentration in high-special-items firm-quarters supports the integration claim: the premium operates where the methodological choices that distinguish the FDPs most affect the reported surprise.

Together, Tables 6–9 document an accuracy premium that is statistically significant, economically meaningful, concentrated where the framework predicts, and—through the predicted surprise—tradeable in real time. The shift from heuristic to accuracy-weighted aggregation is the central empirical signature of Prediction 1. Prices, however, record only the outcome of the

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<sup>29</sup>A third moderator the framework predicts—whether GenAI actually retrieves the firm’s forecasts—requires its own construct, which Section 6 develops. A variant of this table that uses the post-ChatGPT boundary in place of the post-scraping indicator and adds a third partition, a median split on that visibility score, yields similar inferences: the premium concentrates in the high half of each partition (untabulated).



market’s synthesis. Section 6 turns to the inputs: does the premium track the forecasts that GenAI actually retrieves for a firm?

## 6 LLM Visibility and Effective Synthesis Across FDPs

The preceding section establishes that the post-ChatGPT market prices the accuracy-weighted synthesis; the accessibility clause of [Prediction 2](#) adds a condition: the shift should be stronger where LLMs actually retrieve the firm’s forecasts. Integration requires acquisition: the model cannot synthesize forecasts it cannot retrieve. This section introduces *LLM visibility*, a direct measure of that acquisition. For each firm, I ask a browsing-enabled model for the next-quarter consensus EPS and record whether the response contains an EPS figure and how many unique domains it cites; the measure is the EPS-extraction indicator times  $\log(1 + \text{unique cited domains})$ , averaged across ChatGPT and Perplexity responses.<sup>30</sup> It captures what GenAI is *doing*, not what it *can* do. Because the measure is informative only if it amounts to more than repackaged prominence, I show it is distinct from the classical proxies before testing whether acquisition gates the accuracy premium.

*A distinct construct.* Table 10 shows how little LLM visibility shares with the classical prominence proxies: only 47–50 percent of high-visibility firms also sit in the top half of firm size, analyst coverage, or institutional ownership—near the 50 percent coin-flip benchmark for independent median splits—whereas 80 percent of large firms also sit in the top half of analyst coverage. Panels B and C then let the four measures compete: a pooled horse race includes every measure’s high-group triple interaction in one regression, so each measure must explain where the post-scraping accuracy premium concentrates while the others are held in the specification. Against I/B/E/S, LLM visibility is the only measure significant at the five-percent level (5.84,  $p < 0.01$ ); against the equal-weighted benchmark, its coefficient of 3.17 is only marginally significant ( $p < 0.10$ ), as is institutional ownership’s (3.22).

*Acquisition as a precondition.* With distinctness established, a sharper test combines the

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<sup>30</sup>When the browsing-enabled model answers the query, it lists the sources it consulted (e.g., Yahoo Finance, MarketBeat, TipRanks). Counting the unique domains among these citations measures how many distinct forecast sources the model actually retrieves for a firm—the breadth of its information acquisition—and says nothing about the content of any one estimate. [Appendix B](#) reports worked examples.



visibility split with the regime clock: if acquisition gates synthesis, the premium should be larger in high-visibility firms than in low-visibility firms, and that gap should open only once the model can browse. I re-estimate the horse race of Equation (6) within each regime, interacting both surprise measures with an indicator for above-median visibility, and track the high-minus-low gap in the accuracy premium,

$$Premium_r^{\text{High-Low}} = Premium_r^{\text{High vis}} - Premium_r^{\text{Low vis}}, \quad r \in \{R0, R1, R2, R3\}. \quad (7)$$

Table 11 re-estimates the horse race within each regime with year-quarter fixed effects and standard errors two-way clustered by firm and calendar year, as its notes state. In Table 11,  $Premium_r^{\text{High-Low}}$  is statistically insignificant at the five-percent level in the pre-ChatGPT and text-only regimes (R0, R1); the one marginal estimate there, against I/B/E/S in the text-only regime, is negative, a sign the framework does not predict and one the imprecision of that cell counsels against reading. Once the model can browse, the gap emerges: 2.7 to 3.0 in the browsing regime (R2,  $p < 0.05$ ), then 3.1 to 4.2 under native search (R3)—significant against I/B/E/S ( $p < 0.01$ ) but only marginally so against equal-weighting ( $p < 0.10$ ). The premium thus concentrates where forecasts are visible to the model *and* only once it can retrieve them—a pattern consistent with acquisition as a precondition for synthesis rather than an incidental correlate of it. The premium is not confined to high-visibility firms once native search arrives—the low-visibility premium against I/B/E/S is 3.05 ( $p < 0.01$ ) in R3—but the gap between the two groups opens only with retrieval.

*Prominence as an alternative.* The regime timing also weighs against the residual concern that LLM visibility merely proxies for firm prominence. The visibility split is a single firm-level classification applied to every regime, so if high-visibility firms were simply prominent firms, a prominence account would predict a premium gap in the pre-ChatGPT and text-only regimes as well. The gap instead opens only when retrieval arrives, which is difficult to reconcile with an account resting on a static firm characteristic. A prominence effect that itself switched on with browsing would instead surface in the horse race of Table 10, where the classical proxies compete directly for the post-scraping concentration and, against I/B/E/S, none does so at the five-percent level.

The visibility evidence thus delivers what the accessibility clause predicts—an accuracy premium that travels with GenAI’s actual retrieval of forecasts—and narrows the remaining question to the GenAI channel itself, which the final test of Section 7 probes directly.

## 7 Probing the GenAI Channel: ChatGPT Outages as an Exogenous Shock

Sections 4 through 6 establish that the post-ChatGPT market prices an accuracy-weighted synthesis of consensus forecasts and that the resulting accuracy premium emerges only in firms whose forecasts GenAI actually retrieves, and only once it can retrieve them. This section probes the GenAI channel directly by briefly removing it: on announcement days that follow a major ChatGPT outage, the accuracy premium collapses against I/B/E/S but not against the equal-weighted benchmark—the asymmetry a GenAI-mediated channel predicts. An untabulated test of investor search behavior then corroborates the route by which the multi-FDP signal reaches investors.

*Motivation.* A channel that can be interrupted can be identified. To distinguish the GenAI channel from alternative explanations for the post-ChatGPT accuracy premium, I exploit major ChatGPT outages as exogenous, short-duration shocks to GenAI availability, following [Cheng et al. \(2025\)](#) and [Liu and Subasi \(2026\)](#). On announcement dates immediately following a major outage, investors preparing for the announcement have just lost the integration capability that GenAI provides and should revert to the one benchmark that requires no multi-FDP synthesis: the single-FDP I/B/E/S consensus. The prediction is an asymmetric outage effect across the two horse-race benchmarks: against I/B/E/S, the accuracy premium should weaken on these dates as investors substitute toward that fallback; against the equal-weighted benchmark, it should be unchanged, because the accuracy-weighted and equal-weighted aggregates both require multi-FDP synthesis and become equally infeasible when GenAI is down.

*Design.* I collect ChatGPT outage incidents from OpenAI’s public status-page history and define a major outage as any incident lasting at least 240 minutes; this yields four incidents across five unique calendar days through 2024; 2025 announcements therefore enter the post-ChatGPT sample as untreated, which if anything attenuates the estimated collapse. The treatment indicator  $Outage_{i,q}$  equals one if firm  $i$ ’s quarter- $q$  announcement falls on the calendar day immediately following a

major outage and zero otherwise. I re-estimate the horse-race specification of Equation (6) on the post-ChatGPT subsample with  $Outage_{i,q}$  in place of  $Post_q$ , separately for the equal-weighted and I/B/E/S benchmarks, so that  $\beta_3$  and  $\beta_4$  become the outage-day shifts in the two slopes. Because the shock is a single-day treatment, the dependent variable is the day-0 market-adjusted abnormal return rather than the three-day CAR. The five treated days contribute only 84 treated announcements, too few to pin down the level of the outage effect; identification rests instead on the asymmetric sign pattern predicted across the two benchmarks.

*Results.* The estimates in Table 12, Panel B, match the predicted asymmetry. Against the equal-weighted benchmark in column (1), the outage-day change in the accuracy premium is positive but statistically insignificant ( $3.787, t = 1.37$ ); with so few treated days the estimate is imprecise, but it shows no sign of the decline that a reversion toward equal weighting would produce. The column-(1) change combines an insignificant rise in the accuracy-weighted slope ( $\beta_3 = 1.178$ ) with a marginally significant fall in the equal-weighted slope ( $\beta_4 = -2.608, p < 0.10$ ), so what movement there is comes from the equal-weighted loading rather than the accuracy-weighted one. Against the I/B/E/S benchmark in column (2), the outage-day change is  $-3.173$  ( $p < 0.05$ ), combining a marginally significant decline in the accuracy-weighted slope ( $\beta_3 = -1.685, p < 0.10$ ) with a rise in the I/B/E/S slope ( $\beta_4 = 1.488, p < 0.05$ ). Investors reweight toward I/B/E/S in the immediate wake of an outage.

*Corroboration from investor search.* If GenAI substituted for the free distributors it cites, direct Google search for those distributors should fall after ChatGPT; if GenAI instead acts as a gateway that surfaces them, such search should rise. Google search volume is a standard proxy for investor information demand (e.g., Da et al., 2011; Drake et al., 2012). In untabulated platform-day regressions of each platform’s share of total platform-related Google search on group-by-period indicators for 17 platform brands over 2020–2025 (12 free distributors, the four paywalled FDPs, and Zacks as the free FDP), the post-ChatGPT share rises by 2.1 percentage points for free distributors and falls by 7.5 and 7.1 percentage points for the free and paywalled FDPs, respectively, and the pattern is stable across the three GPT-capability regimes. The evidence is aggregate and

associational, but its sign fits the gateway reading: the multi-FDP consensus information the premium requires is reaching investors.

The asymmetry across the two benchmarks is the identifying signature of the GenAI channel. A shock that removes multi-FDP synthesis should leave the contrast between two synthesized aggregates untouched and move only the contrast with the single-FDP fallback, and that is what the estimates show: the accuracy premium collapses against the I/B/E/S benchmark while the equal-weighted contrast shows no significant change. This sign pattern is difficult to generate without GenAI-mediated multi-FDP integration. Together with the regime and cross-sectional evidence in Section 5, the visibility evidence in Section 6, and the gateway evidence above, the outage test points to GenAI as the channel through which the post-ChatGPT accuracy-weighted synthesis reaches market prices.

## 8 Conclusion

This paper shows that after the release of ChatGPT, market reactions to earnings announcements increasingly reflect an accuracy-weighted synthesis across the five major forecast data providers, departing from the equal-weighted or single-FDP heuristics that characterized the pre-ChatGPT market. Three pieces of evidence support this conclusion. First, the post-ChatGPT market discriminates across FDPs by accuracy: the within-firm-quarter ERC gradient in provider accuracy rank is roughly twice as steep after ChatGPT as before, and the firm-level response to the I/B/E/S consensus surprise weakens where I/B/E/S has lost relative accuracy but not where it has gained. The same selectivity extends to the consensus's predictive value: the ERC gradient on a real-time measure of how closely the consensus has anticipated the firm's year-ahead street numbers roughly doubles after ChatGPT.

Second, after ChatGPT the accuracy-weighted multi-provider consensus commands a statistically and economically significant accuracy premium over both the equal-weighted alternative and the I/B/E/S-only consensus. The premium concentrates where the integration burden is heaviest (large special items) and, against I/B/E/S, where GenAI synthesis has the most discriminating work to do (high cross-FDP disagreement). It is also tradeable: a quintile long-short

portfolio sorted on the predicted surprise—the ex ante gap between the accuracy-weighted and each heuristic consensus—earns 0.63–1.35 percent per announcement. The premium also tracks the enabling technology: it climbs across GPT-capability regimes, reaching 4.91 percentage points against I/B/E/S under native search, and it emerges in firms whose forecasts the model demonstrably retrieves (LLM visibility)—but only once retrieval is possible.

The third piece probes the GenAI channel itself: major ChatGPT outages produce the asymmetry a GenAI-mediated multi-FDP channel predicts. On announcement days that follow an outage, the accuracy premium against I/B/E/S collapses, while the equal-weighted contrast shows no significant change. An untabulated test of post-ChatGPT search behavior corroborates the acquisition route: investor search shifts toward the free distributors GenAI cites, consistent with GenAI broadening, rather than substituting for, direct access to cross-FDP consensus information.

These findings make three contributions. First, they reposition the choice of forecast data provider from a researcher’s measurement decision to an endogenous market outcome that responds to the information environment, extending recent work on cross-FDP differences in consensus information ([Larocque et al., 2025](#)). The historical reliance on I/B/E/S reflects an equilibrium that held only while cross-provider integration was costly; for measurement practice, the benchmark against which earnings surprises should be gauged shifts with the technology investors use to read the consensus. Second, they identify a specific channel through which GenAI reaches capital markets: low-cost multi-source synthesis of analyst forecasts, which the market now prices into earnings-announcement reactions. That specificity isolates the analyst-information channel from the broader set of GenAI effects on investor behavior and asset prices documented in recent work (e.g., [Bertomeu et al., 2025](#); [Cheng et al., 2025](#); [Liu and Subasi, 2026](#)). Third, they sharpen the disclosure-processing-cost framework of [Blankespoor et al. \(2020\)](#) by pinpointing cross-source integration—distinct from awareness, acquisition, or integration within a single source (e.g., [Blankespoor et al., 2014](#); [Blankespoor, 2019](#); [Drake et al., 2015](#))—as the cost component that prior single-source technologies left intact and that GenAI uniquely relaxes.

Four caveats bound these claims. The accuracy-weighted construction is a tractable proxy: the

evidence establishes that market reactions align more closely with an accuracy-sensitive synthesis than with the heuristic benchmarks, not that the market applies these particular weights. The LLM visibility measure is a point-in-time snapshot of inherently variable model outputs and has limited reliability (e.g., [Chen et al., 2023](#); [Lopez-Lira et al., 2025](#)). The predictive-value measure of Section 4 rewards internal consistency of street-earnings rules alongside genuine foresight, so only its post-ChatGPT increment is identified, and that increment concerns the consensus common to all providers rather than provider selection. And the outage test rests on few treated days, so identification comes from the asymmetric sign across benchmarks rather than from level estimates, which are too imprecise to bear a magnitude interpretation.

Several questions remain open. First, only a longer post-ChatGPT panel can settle whether the accuracy premium widens or contracts as GenAI tools evolve and as institutional investors embed them in their workflows. Second, the evidence covers EPS forecasts alone; applying the same design to revenue, cash-flow, or industry-specific forecasts would show whether the cross-FDP synthesis advantage extends beyond earnings. Third, the analysis treats FDPs as fixed information producers, but the market's new discrimination across them gives providers an incentive to revisit their methodologies; such a supply-side response would bear on both the disclosure-processing-cost framework and the structure of the FDP industry.

The broader lesson is that the market's earnings benchmark is technology-bound. For decades, a single provider's consensus stood in for the market's expectation because integrating across providers was costly; GenAI has relaxed that constraint, and the benchmark has moved. The market now weights providers by the accuracy they have earned, not the convenience of reaching them—what investors can cheaply integrate shapes what prices reflect.

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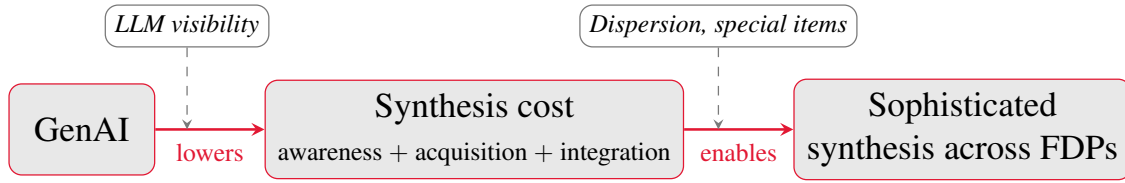


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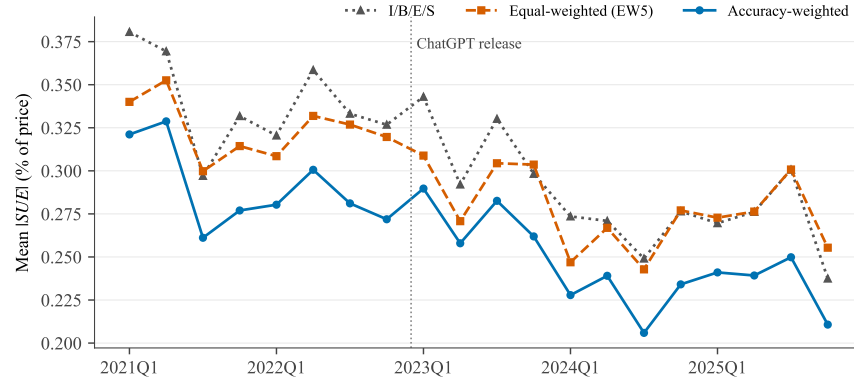
**FIGURE 1. Theoretical Framework**



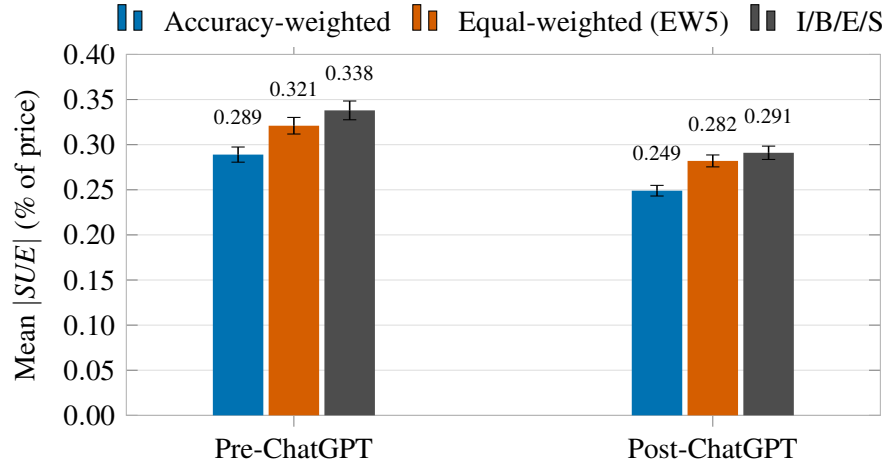
*Notes.* This figure summarizes the theoretical framework. Generative AI (GenAI) lowers the cost of synthesizing analyst consensus forecasts across the five forecast data providers (FDPs), the joint cost of *awareness*, *acquisition*, and *integration* in the Blankespoor et al. (2020) decomposition, which in turn enables the market to price a sophisticated, accuracy-weighted synthesis across providers in place of a single-FDP or equal-weighted heuristic. The strength of the channel is moderated by how *accessible* synthesis is for a given firm (LLM visibility, the information-acquisition channel) and by how *valuable* it is (cross-FDP disagreement and special-items magnitude). These mechanisms map to Predictions 1 and 2, developed in Section 2; Prediction 3 extends the shift to a second dimension of consensus quality: the consensus’s predictive value for the firm’s future street numbers.

**FIGURE 2. Mean Magnitude of  $SUE$ : Accuracy-Weighted vs. Heuristic Consensus Benchmarks**

**(a) Quarterly mean  $SUE$  by consensus method**

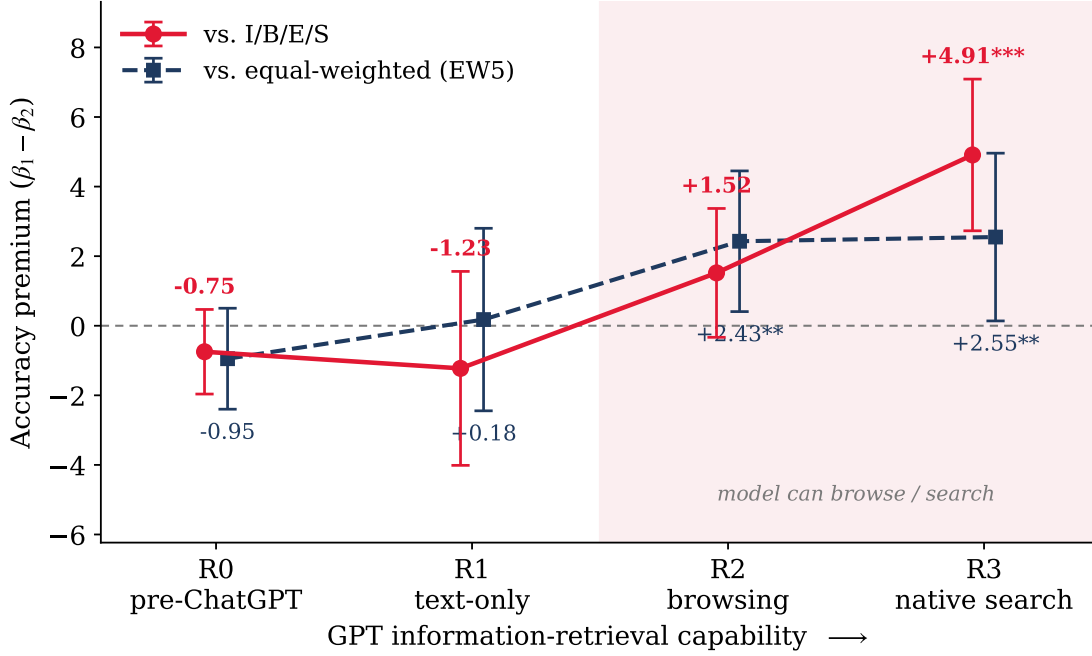


**(b) Mean  $SUE$  by consensus method**



*Notes.* This figure compares the accuracy of the accuracy-weighted multi-vendor consensus  $SUE^{Acc}$  against the equal-weighted five-FDP consensus  $SUE^{EW5}$  and the I/B/E/S consensus  $SUE^{IBES}$  across the five-FDP sample (FactSet, Bloomberg, Zacks, Capital IQ, I/B/E/S) of S&P 1500 firm-quarters with announcement dates in 2021Q1–2025Q4.  $|SUE|$  is the absolute standardized unexpected earnings scaled as a percentage of pre-announcement price, winsorized at the (1, 99) level. **Panel (a)** plots the quarterly mean  $|SUE|$  for each of the three methods over 20 announcement quarters. The vertical reference line marks the public release of ChatGPT (November 30, 2022). **Panel (b)** shows the pooled mean  $|SUE|$  within each Pre- vs. Post-ChatGPT regime, with 95% confidence intervals from the cell-level standard error of the mean; the y-axis starts at zero so that bar height is proportional to the mean.

**FIGURE 3. The Accuracy Premium Rises with GPT Retrieval Capability**



*Notes.* This figure plots the *accuracy premium*, the contrast  $\beta_1 - \beta_2$  between the loading of the three-day market-adjusted  $CAR_{[-1,+1]}$  (in percent) on the accuracy-weighted consensus surprise  $SUE^{Acc}$  and its loading on the benchmark surprise (both in percent of price), estimated separately within each GPT-capability regime; the vertical axis is therefore in percentage points of announcement return per one-percentage-point surprise. The estimates are taken directly from [Table 8](#): the red line uses the I/B/E/S benchmark (Panel B) and the navy line the equal-weighted five-FDP benchmark (Panel A). Regimes are ordered by ChatGPT's information-retrieval capability: **R0** (before 2022-11-30, pre-ChatGPT), **R1** (text-only), **R2** (browsing, from 2023-09-27), and **R3** (native search, from 2024-10-31); the shaded region marks the regimes in which the model can retrieve forecasts from the web. Whiskers are 95% confidence intervals; point labels report the premium and its significance (\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ ) from firm-clustered standard errors. The premium is statistically indistinguishable from zero before retrieval is available, turns positive once the model can browse, and grows monotonically thereafter, the dose-response [Prediction 2](#) predicts.

**TABLE 1. Sample Selection**

| <b>Sample construction</b> (firm $\times$ quarter)                                                                         | <b>Observations</b> | <b># Firms</b> |
|----------------------------------------------------------------------------------------------------------------------------|---------------------|----------------|
| S&P 1500 firm-quarters with an earnings announcement<br>(announcement date 2021–2025)                                      | 29,395              | 1,502          |
| Retain firm-quarters with non-missing consensus from all five<br>FDPs (FactSet, Bloomberg, Zacks, Capital IQ, and I/B/E/S) | 25,027              | 1,353          |
| Drop firm-quarters missing market-outcome variables or<br>required controls                                                | (1,179)             | (13)           |
| Drop firm-quarters missing any constructed return or<br>predicted-surprise variable                                        | (10)                | (1)            |
| Drop firms with only one firm-quarter (no within-firm variation<br>for firm fixed effects)                                 | (5)                 | (5)            |
| <b>Final Sample</b>                                                                                                        | <b>23,833</b>       | <b>1,334</b>   |

*Notes.* This table reports the sample selection procedure for the firm-quarter sample. Counts in parentheses are the firm-quarters (firms) dropped at that step; all other rows report the number surviving. The sample is used in the main across-FDP tests and is built from CRSP, Compustat, I/B/E/S, and the five-FDP consensus panel.



**TABLE 2. Descriptive Statistics**

|                                                                    | N      | Mean   | SD    | Min     | p25    | Median | p75    | Max    |
|--------------------------------------------------------------------|--------|--------|-------|---------|--------|--------|--------|--------|
| <b>Dependent Variables: Abnormal Returns (%)</b>                   |        |        |       |         |        |        |        |        |
| $CAR_{[-1,+1]}^{Mkt}$                                              | 23,833 | 0.204  | 7.739 | -22.532 | -3.933 | 0.168  | 4.255  | 23.535 |
| $CAR_{[-1,+1]}^{FFC}$                                              | 23,833 | 0.196  | 7.635 | -22.779 | -3.776 | 0.186  | 4.135  | 23.606 |
| $BHAR_{[-2,+1]}^{Mkt}$                                             | 23,833 | 0.228  | 7.978 | -22.803 | -4.165 | 0.177  | 4.418  | 24.992 |
| $BHAR_{[-2,+1]}^{FFC}$                                             | 23,833 | 0.193  | 7.847 | -22.935 | -3.966 | 0.111  | 4.255  | 24.819 |
| <b>Per-FDP Standardized Earnings Surprise (SUE<sup>g</sup>, %)</b> |        |        |       |         |        |        |        |        |
| $SUE^{FactSet}$                                                    | 23,833 | 0.132  | 0.534 | -2.190  | 0.000  | 0.074  | 0.238  | 2.484  |
| $SUE^{Bloomberg}$                                                  | 23,833 | -0.222 | 1.659 | -10.049 | -0.362 | -0.027 | 0.162  | 5.780  |
| $SUE^{Zacks}$                                                      | 23,833 | 0.149  | 0.467 | -1.606  | 0.000  | 0.075  | 0.240  | 2.291  |
| $SUE^{CapIQ}$                                                      | 23,833 | 0.137  | 0.530 | -2.185  | 0.001  | 0.076  | 0.243  | 2.477  |
| $SUE^{IBES}$                                                       | 23,833 | 0.130  | 0.548 | -2.271  | 0.000  | 0.075  | 0.241  | 2.502  |
| <b>Cross-FDP Composites and Predicted Surprise</b>                 |        |        |       |         |        |        |        |        |
| $SUE^{Acc}$                                                        | 23,833 | 0.110  | 0.450 | -1.656  | -0.015 | 0.059  | 0.212  | 1.972  |
| $SUE^{EW5}$                                                        | 23,833 | 0.066  | 0.510 | -1.936  | -0.064 | 0.041  | 0.201  | 1.942  |
| $Standardized\ Pred.\ Surp.^{EW5}$                                 | 23,833 | 0.606  | 0.116 | 0.000   | 0.590  | 0.619  | 0.631  | 1.000  |
| $Standardized\ Pred.\ Surp.^{IBES}$                                | 23,833 | 0.483  | 0.109 | 0.000   | 0.469  | 0.478  | 0.493  | 1.000  |
| <b>Firm and Event Controls</b>                                     |        |        |       |         |        |        |        |        |
| <i>Log Market Equity</i>                                           | 23,833 | 8.992  | 1.448 | 6.390   | 7.923  | 8.769  | 9.942  | 12.883 |
| <i>Book-to-Market</i>                                              | 23,833 | 0.440  | 0.361 | -0.159  | 0.170  | 0.362  | 0.634  | 1.670  |
| <i>Leverage</i>                                                    | 23,833 | 0.314  | 0.218 | 0.001   | 0.133  | 0.301  | 0.444  | 1.008  |
| <i>Prior 12-Month Return</i>                                       | 23,833 | 0.201  | 0.487 | -0.589  | -0.101 | 0.112  | 0.379  | 2.400  |
| <i>Log Price</i>                                                   | 23,833 | 4.166  | 0.980 | 1.880   | 3.491  | 4.178  | 4.831  | 6.580  |
| <i>Return Volatility</i>                                           | 23,833 | 0.021  | 0.009 | 0.009   | 0.015  | 0.019  | 0.025  | 0.054  |
| <i>Analyst Dispersion</i>                                          | 23,833 | 0.002  | 0.003 | 0.000   | 0.000  | 0.001  | 0.002  | 0.018  |
| <i>Busyness</i>                                                    | 23,833 | 4.944  | 0.973 | 1.946   | 4.431  | 5.124  | 5.684  | 6.299  |
| <i> Δ Income Before Extraordinary Items </i>                       | 23,833 | 0.012  | 0.026 | 0.000   | 0.001  | 0.004  | 0.011  | 0.178  |
| <i> Δ Shares Outstanding </i>                                      | 23,833 | 0.010  | 0.021 | 0.000   | 0.001  | 0.003  | 0.009  | 0.150  |
| <i> Special Items </i>                                             | 23,833 | 0.003  | 0.010 | 0.000   | 0.000  | 0.000  | 0.002  | 0.074  |
| <i>High Stock-Based Compensation</i>                               | 23,833 | 0.500  | 0.500 | 0.000   | 0.000  | 0.000  | 1.000  | 1.000  |
| <i>Unexpected Item</i>                                             | 23,833 | 0.686  | 0.464 | 0.000   | 0.000  | 1.000  | 1.000  | 1.000  |
| <i>Stock Split</i>                                                 | 23,833 | 0.003  | 0.051 | 0.000   | 0.000  | 0.000  | 0.000  | 1.000  |
| <i>Fiscal Q4</i>                                                   | 23,833 | 0.250  | 0.433 | 0.000   | 0.000  | 0.000  | 0.000  | 1.000  |
| <i>Institutional Ownership</i>                                     | 23,833 | 0.850  | 0.238 | 0.011   | 0.813  | 0.940  | 1.000  | 1.000  |
| <i># Analysts</i>                                                  | 23,833 | 10.914 | 6.536 | 2.000   | 6.000  | 10.000 | 15.000 | 31.000 |
| <b>Cross-Sectional Partitioning Indicators</b>                     |        |        |       |         |        |        |        |        |
| <i>High Visibility</i>                                             | 23,827 | 0.489  | 0.500 | 0.000   | 0.000  | 0.000  | 1.000  | 1.000  |

(continued on next page)

|                                   | <b>N</b> | <b>Mean</b> | <b>SD</b> | <b>Min</b> | <b>p25</b> | <b>Median</b> | <b>p75</b> | <b>Max</b> |
|-----------------------------------|----------|-------------|-----------|------------|------------|---------------|------------|------------|
| <i>High Cross-FDP Dispersion</i>  | 23,833   | 0.500       | 0.500     | 0.000      | 0.000      | 0.000         | 1.000      | 1.000      |
| <i>High Special Items</i>         | 23,833   | 0.500       | 0.500     | 0.000      | 0.000      | 0.000         | 1.000      | 1.000      |
| <i>IBES Accuracy-Rank Up</i>      | 23,833   | 0.099       | 0.299     | 0.000      | 0.000      | 0.000         | 0.000      | 1.000      |
| <i>IBES Accuracy-Rank Down</i>    | 23,833   | 0.059       | 0.236     | 0.000      | 0.000      | 0.000         | 0.000      | 1.000      |
| <i>ChatGPT Outage (prior day)</i> | 23,833   | 0.004       | 0.059     | 0.000      | 0.000      | 0.000         | 0.000      | 1.000      |

*Notes.* This table reports descriptive statistics for the two analytic samples. This table reports the across-FDP sample of S&P 1500 firm-quarters with announcement dates in 2021–2025 used in the main tests. All continuous variables are winsorized at the (1, 99) level. *High Visibility* is defined for 23,827 of the 23,833 firm-quarters. *# Analysts* enters the regressions in logs. Variable definitions are in [Appendix A](#).

**TABLE 3. How Often Do FDPs’ Street Actuals Disagree?**

|            | I/B/E/S | FactSet | Bloomberg | Zacks |
|------------|---------|---------|-----------|-------|
| FactSet    | 20.8    |         |           |       |
| Bloomberg  | 87.9    | 87.9    |           |       |
| Zacks      | 33.5    | 37.3    | 90.4      |       |
| Capital IQ | 14.4    | 19.0    | 88.1      | 31.2  |

*Notes.* All five providers report the same street actual in only 6.8 percent of the 23,833 sample firm-quarters, and the largest cross-provider gap is at least five cents per share in 77.1 percent. Each cell reports the percentage of the 23,833 sample firm-quarters in which the row and column providers publish different street actuals for the same firm-quarter, where “different” means the two actuals differ by at least half a cent per share; all five actuals are non-missing for every sample firm-quarter. Because both providers observe the same GAAP filing, any difference in their reported actuals reflects the providers’ own construction rules for street earnings—the treatment of unexpected charges and gains, stock-based compensation, and other adjustments ([Bochkay et al., 2022](#))—rather than differences in the underlying analyst pool. Disagreement at the actuals level implies disagreement in the earnings surprise even for identical forecasts.

**TABLE 4. Initial Evidence on Selective Reliance**

**Panel A. Earnings-response coefficients by forecast-provider accuracy rank**

| Provider                                      | Pre-ChatGPT                      |                                   | Post-ChatGPT                     |                                    |
|-----------------------------------------------|----------------------------------|-----------------------------------|----------------------------------|------------------------------------|
|                                               | $\beta$                          | $\gamma$                          | $\beta$                          | $\gamma$                           |
| I/B/E/S                                       | 2.591***<br>(6.60)               | +0.022<br>(0.17)                  | 2.611***<br>(5.49)               | +0.295**<br>(2.09)                 |
| FactSet                                       | 2.316***<br>(4.78)               | +0.050<br>(0.36)                  | 2.059***<br>(3.80)               | +0.380**<br>(2.42)                 |
| Bloomberg                                     | 0.985***<br>(5.82)               | -0.058<br>(-0.80)                 | 0.789***<br>(5.01)               | +0.162**<br>(2.10)                 |
| Zacks                                         | 3.603***<br>(8.35)               | -0.139<br>(-1.04)                 | 4.757***<br>(8.57)               | -0.020<br>(-0.13)                  |
| Capital IQ                                    | 1.107**<br>(2.40)                | +0.543***<br>(3.94)               | 2.483***<br>(4.94)               | +0.349**<br>(2.40)                 |
| <b>Stacked</b>                                | <b>1.206***</b><br><b>(9.21)</b> | <b>+0.306***</b><br><b>(6.93)</b> | <b>0.973***</b><br><b>(8.09)</b> | <b>+0.621***</b><br><b>(11.65)</b> |
| <i>Observations</i>                           | 47,130                           |                                   | 72,035                           |                                    |
| <b>Stacked <math>\Delta</math> (Post-Pre)</b> |                                  |                                   | <b>-0.232</b><br><b>(-1.38)</b>  | <b>+0.316***</b><br><b>(4.91)</b>  |

**Panel B. Heterogeneous post-ChatGPT effects by firm-level I/B/E/S accuracy-rank change**

| Dep. Var.:                                | Market-adjusted $CAR_{[-1,+1]}$   |                                   | FFC $CAR_{[-1,+1]}$                |                                   |
|-------------------------------------------|-----------------------------------|-----------------------------------|------------------------------------|-----------------------------------|
|                                           | (1)                               | (2)                               | (3)                                | (4)                               |
| $SUE^{IBES}$                              | 2.859***<br>(13.91)               | 2.845***<br>(14.03)               | 2.775***<br>(13.67)                | 2.776***<br>(13.84)               |
| $SUE^{IBES} \times Post$                  | 1.029***<br>(3.56)                | 0.978***<br>(3.32)                | 1.136***<br>(4.11)                 | 1.065***<br>(3.76)                |
| $SUE^{IBES} \times IBES Up$               | 0.693<br>(1.24)                   | 0.628<br>(1.16)                   | 0.803<br>(1.46)                    | 0.737<br>(1.39)                   |
| $SUE^{IBES} \times IBES Down$             | 1.397*<br>(1.81)                  | 1.582**<br>(2.00)                 | 1.913***<br>(2.61)                 | 2.080***<br>(2.74)                |
| $SUE^{IBES} \times Post \times IBES Up$   | <b>-1.248</b><br><b>(-1.44)</b>   | <b>-1.135</b><br><b>(-1.30)</b>   | <b>-1.291</b><br><b>(-1.40)</b>    | <b>-1.185</b><br><b>(-1.26)</b>   |
| $SUE^{IBES} \times Post \times IBES Down$ | <b>-2.278**</b><br><b>(-2.27)</b> | <b>-2.344**</b><br><b>(-2.15)</b> | <b>-2.758***</b><br><b>(-2.70)</b> | <b>-2.791**</b><br><b>(-2.51)</b> |
| $Post \times IBES Up$                     | 0.353<br>(1.13)                   | 0.232<br>(0.68)                   | 0.437<br>(1.37)                    | 0.297<br>(0.85)                   |
| $Post \times IBES Down$                   | 0.544<br>(1.34)                   | 0.713<br>(1.61)                   | 0.678*<br>(1.68)                   | 0.859*<br>(1.95)                  |
| Controls                                  | No                                | Yes                               | No                                 | Yes                               |
| Firm FE                                   | Yes                               | Yes                               | Yes                                | Yes                               |
| Year FE                                   | Yes                               | Yes                               | Yes                                | Yes                               |
| Observations                              | 23,833                            | 23,833                            | 23,833                             | 23,833                            |
| Adjusted $R^2$                            | 0.052                             | 0.070                             | 0.054                              | 0.072                             |

*Notes.* Both panels test whether the market relies more on more accurate forecasts. **Panel A:** within each firm-quarter the five FDPs are ranked by their ex-ante trailing four-quarter mean absolute forecast error (MAFE; Rank 1 = least accurate, Rank 5 = most accurate); pooling across firm-quarters,  $CAR_{[-1,+1]}^{Mkt} =$

$\alpha + \beta SUE^g + \delta Rank^g + \gamma(SUE^g \times Rank^g)$ , estimated without controls or fixed effects, where  $\gamma$  is the ERC increment per one-rank accuracy improvement. Each provider row is its own regression; the *Stacked* row pools all five FDPs; the *Stacked  $\Delta$*  row reports the  $SUE \times Post$  and  $SUE \times Rank \times Post$  coefficients from the stacked regression that pools both periods and interacts every term with *Post*. **Panel B:** firms are split by the change in IBES's rounded mean rank among the five FDPs between the pre- and post-ChatGPT windows ( $IBESUp = 1$  if the rank improved by  $\geq 2$  positions;  $IBESDown = 1$  if it worsened by  $\geq 2$ ); the bolded triple interactions are the key coefficients; all lower-order terms are included but not reported. *Controls* are the firm and event controls defined in [Appendix A](#), and Panel B includes firm and year fixed effects. Pre-ChatGPT (Post-ChatGPT) covers announcements before (on or after) 2022-11-30. Standard errors are clustered by firm; continuous variables are winsorized at (1, 99). *t*-statistics in parentheses; \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Variable definitions are in [Appendix A](#).

**TABLE 5. The Predictive Value of the I/B/E/S Consensus: Measurement and Post-ChatGPT Pricing**

| Dep. Var.:                                                                | Single post-ChatGPT indicator    |                                  | GPT-capability regimes           |                                  |
|---------------------------------------------------------------------------|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
|                                                                           | Mkt CAR<br>(1)                   | FFC CAR<br>(2)                   | Mkt CAR<br>(3)                   | FFC CAR<br>(4)                   |
| $SUE^{IBES} (\beta_1)$                                                    | 2.237***<br>(8.71)               | 2.241***<br>(9.02)               | 2.237***<br>(8.74)               | 2.241***<br>(9.05)               |
| $SUE^{IBES} \times Post (\beta_2)$                                        | 0.534<br>(1.54)                  | 0.565*<br>(1.67)                 |                                  |                                  |
| $SUE^{IBES} \times \text{Regime 1 (text-only)}$                           |                                  |                                  | 0.868*<br>(1.91)                 | 1.016**<br>(2.29)                |
| $SUE^{IBES} \times \text{Regime 2 (browsing)}$                            |                                  |                                  | 0.785<br>(1.62)                  | 0.591<br>(1.18)                  |
| $SUE^{IBES} \times \text{Regime 3 (native search)}$                       |                                  |                                  | 0.162<br>(0.33)                  | 0.268<br>(0.57)                  |
| $SUE^{IBES} \times \text{Predictive Value}^{IBES} (\beta_3)$              | 2.172***<br>(4.03)               | 2.119***<br>(4.11)               | 2.174***<br>(4.05)               | 2.125***<br>(4.13)               |
| $SUE^{IBES} \times \text{Predictive Value}^{IBES} \times Post (\beta_4)$  | <b>2.049***</b><br><b>(2.68)</b> | <b>2.098***</b><br><b>(2.82)</b> |                                  |                                  |
| $SUE^{IBES} \times \text{Predictive Value}^{IBES} \times \text{Regime 1}$ |                                  |                                  | 0.919<br>(0.93)                  | 0.941<br>(0.97)                  |
| $SUE^{IBES} \times \text{Predictive Value}^{IBES} \times \text{Regime 2}$ |                                  |                                  | <b>1.435</b><br><b>(1.45)</b>    | <b>1.777*</b><br><b>(1.80)</b>   |
| $SUE^{IBES} \times \text{Predictive Value}^{IBES} \times \text{Regime 3}$ |                                  |                                  | <b>3.540***</b><br><b>(3.08)</b> | <b>3.365***</b><br><b>(3.02)</b> |
| Lower-order terms / Controls                                              | Yes                              | Yes                              | Yes                              | Yes                              |
| Firm FE / Year FE                                                         | Yes                              | Yes                              | Yes                              | Yes                              |
| Observations                                                              | 22,356                           | 22,356                           | 22,356                           | 22,356                           |
| Adjusted $R^2$                                                            | 0.082                            | 0.083                            | 0.082                            | 0.083                            |

*Notes.* This table tests whether the post-ChatGPT strengthening of announcement-window reactions concentrates in firms whose I/B/E/S consensus has closely anticipated the firm's future street numbers.  $Predictive\ Value_{i,q}^{IBES} = 1 - error_{i,q}$ , where  $error_{i,q}$  is the consensus's trailing relative anticipation error: the mean own-basis distance  $|actual_{i,q+4} - consensus_{i,q}|$  over up to the twenty most recent source quarters whose year-ahead actuals are public at the announcement (at least twelve required; a stock-split guard applies), scaled by the window mean  $|actual|$  so the measure is comparable across firms, and winsorized at (1, 99). The measure is continuous (mean 0.56, standard deviation 0.30) and observable in real time; higher values mean the consensus has anticipated the firm's year-ahead street numbers more closely. It covers 22,364 of the 23,833 firm-quarters (8,730 pre-ChatGPT), of which 22,356 enter the regressions; the uncovered firm-quarters are recent listings with fewer than twelve scorable history quarters.  $SUE^{IBES}$  is the I/B/E/S consensus surprise. Columns (1)–(2) regress the three-day market-adjusted and Fama-French-Carhart four-factor  $CAR_{[-1,+1]}$  on  $SUE^{IBES}$ ,  $Predictive\ Value^{IBES}$ ,  $Post$  (one for announcements on or after November 30, 2022), and all their two- and three-way interactions;  $\beta_3$  is the pre-ChatGPT gradient of the ERC in predictive value, and the bolded  $\beta_4$  is its post-ChatGPT increment. Columns (3)–(4) replace the single  $Post$  indicator with the three GPT-capability regimes of [Appendix A](#) (text-only from November 30, 2022; browsing from September 27, 2023; native search

from October 31, 2024); because the native-search regime spans only the final five sample quarters, the regime profile is descriptive and the pooled increment in columns (1)–(2) is the formal test. All columns include the lower-order terms, the Table 7 controls (including dispersion, volatility, and the unexpected-item flag), and firm and year fixed effects; standard errors are clustered by firm.  $t$ -statistics in parentheses;  $*p < 0.10$ ,  $**p < 0.05$ ,  $***p < 0.01$ .

**TABLE 6. Accuracy Persistence and the Predicted-Surprise Trading Strategy****Panel A. Persistence of FDP accuracy ranks (trailing-MAFE rank, quarter  $t$  to  $t + 1$ )**

| $rank_t$  | $\Pr(rank_{t+1} = j \mid rank_t = i), \text{ in } \%$ |             |             |             |                        |
|-----------|-------------------------------------------------------|-------------|-------------|-------------|------------------------|
|           | $j = 1 \text{ (least)}$                               | $j = 2$     | $j = 3$     | $j = 4$     | $j = 5 \text{ (most)}$ |
| 1 (least) | <b>76.1</b>                                           | 12.1        | 4.9         | 3.4         | 3.6                    |
| 2         | 12.1                                                  | <b>54.1</b> | 19.7        | 8.9         | 5.2                    |
| 3         | 4.7                                                   | 19.5        | <b>46.7</b> | 21.1        | 8.0                    |
| 4         | 3.4                                                   | 9.1         | 20.8        | <b>48.3</b> | 18.5                   |
| 5 (most)  | 3.8                                                   | 5.2         | 7.9         | 18.4        | <b>64.7</b>            |

Spearman  $\rho = 0.71$ ; overall same-rank rate = 58%; 109,915 quarter-to-quarter pairs.

**Panel B. Trading strategy from predicted surprise**

| Dep. Var.:                       | Market-adj. $BHAR_{[-2,+1]}$ |                           | FFC $BHAR_{[-2,+1]}$      |                           |
|----------------------------------|------------------------------|---------------------------|---------------------------|---------------------------|
| Benchmark:                       | Equal-Weighted               | IBES                      | Equal-Weighted            | IBES                      |
|                                  | (1)                          | (2)                       | (3)                       | (4)                       |
| <i>Standardized Pred. Surp.</i>  | <b>2.493***</b><br>(4.10)    | <b>3.433***</b><br>(5.35) | <b>2.440***</b><br>(4.08) | <b>3.357***</b><br>(5.25) |
| <b>Long-Short Return (Q5-Q1)</b> | <b>+0.73%***</b>             | <b>+1.35%***</b>          | <b>+0.63%***</b>          | <b>+1.35%***</b>          |
| Controls / Firm FE / Year FE     | Yes                          | Yes                       | Yes                       | Yes                       |
| Observations (firm-quarters)     | 23,833                       | 23,833                    | 23,833                    | 23,833                    |
| Adjusted $R^2$                   | 0.016                        | 0.017                     | 0.018                     | 0.019                     |

*Notes.* This table motivates the accuracy-weighted consensus. **Panel A** reports the quarter-to-quarter persistence of forecast-provider accuracy ranks. Within each firm-quarter the five FDPs are ranked by their trailing four-quarter mean absolute forecast error (Rank 1 = least accurate, Rank 5 = most accurate, the same ex-ante measure used to build  $SUE^{Acc}$ ); cell  $(i, j)$  is the share of provider-quarters at Rank  $i$  that move to Rank  $j$  the following quarter (rows sum to 100 up to rounding). **Panel B** tests whether each of two predicted-surprise signals forecasts the four-day announcement-window abnormal return: for benchmark  $X \in \{EW5, IBES\}$  the predicted surprise is  $(Consensus^{Acc} - Consensus^X)/\text{price}$ , min-max standardized to  $[0, 1]$ ; columns (1)–(2) use the market-adjusted  $BHAR_{[-2,+1]}$  and (3)–(4) the Fama-French-Carhart (FFC) four-factor  $BHAR_{[-2,+1]}$ . The Long-Short row reports the abnormal return, in percent per announcement, from going long the top quintile and short the bottom quintile of predicted surprise and holding each position over the  $[-2, +1]$  window. All Panel B columns include firm and year fixed effects, the firm and event controls defined in [Appendix A](#), and firm-clustered standard errors; continuous variables are winsorized at  $(1, 99)$ .  $t$ -statistics in parentheses; \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Variable definitions are in [Appendix A](#).



**TABLE 7. The Accuracy Premium: Horse Races Between the Accuracy-Weighted and Heuristic Surprises**

**Panel A. Horse race:  $SUE^{Acc}$  vs  $SUE^{EW5}$ .**

| Dep. Var.:                                    | Mkt CAR <sub>[-1,+1]</sub> |                      |                            | FFC CAR <sub>[-1,+1]</sub> |                      |                            |
|-----------------------------------------------|----------------------------|----------------------|----------------------------|----------------------------|----------------------|----------------------------|
|                                               | (1)<br>Pre                 | (2)<br>Post          | (3)<br>Pooled              | (4)<br>Pre                 | (5)<br>Post          | (6)<br>Pooled              |
| $SUE^{Acc} (\beta_1)$                         | 1.376***<br>(3.45)         | 3.420***<br>(9.38)   | 1.307***<br>(3.60)         | 1.401***<br>(3.63)         | 3.519***<br>(10.01)  | 1.290***<br>(3.67)         |
| $SUE^{Acc} \times Post (\beta_3)$             |                            |                      | 2.003***<br>(4.08)         |                            |                      | 2.135***<br>(4.47)         |
| $SUE^{EW5} (\beta_2)$                         | 2.324***<br>(6.10)         | 1.767***<br>(5.56)   | 2.279***<br>(6.84)         | 2.379***<br>(6.44)         | 1.734***<br>(5.62)   | 2.284***<br>(7.03)         |
| $SUE^{EW5} \times Post (\beta_4)$             |                            |                      | -0.399<br>(-0.92)          |                            |                      | -0.449<br>(-1.06)          |
| Premium (Pre-ChatGPT)                         | -0.948<br>(-1.28)          |                      | -0.972<br>(-1.47)          | -0.978<br>(-1.37)          |                      | -0.994<br>(-1.55)          |
| Premium (Post-ChatGPT)                        |                            | 1.654***<br>(2.62)   | 1.429**<br>(2.39)          |                            | 1.785***<br>(2.93)   | 1.590***<br>(2.74)         |
| <b><math>\Delta</math> Premium (Post-Pre)</b> |                            |                      | <b>2.401***<br/>(2.75)</b> |                            |                      | <b>2.584***<br/>(3.03)</b> |
| <b>Control Variables</b>                      |                            |                      |                            |                            |                      |                            |
| <i>Log Market Equity</i>                      | -5.348***<br>(-5.35)       | -4.964***<br>(-5.35) | -2.848***<br>(-6.00)       | -5.585***<br>(-5.59)       | -5.125***<br>(-5.74) | -3.238***<br>(-7.19)       |
| <i>Book-to-Market</i>                         | 0.413<br>(0.36)            | -1.315<br>(-1.22)    | 0.168<br>(0.28)            | 0.849<br>(0.75)            | -1.245<br>(-1.16)    | 0.309<br>(0.53)            |
| <i>Leverage</i>                               | -4.957**<br>(-2.38)        | -3.652*<br>(-1.70)   | -0.913<br>(-0.76)          | -4.561**<br>(-2.17)        | -4.067**<br>(-1.97)  | -1.312<br>(-1.13)          |
| <i>Prior 12-Month Return</i>                  | -0.295<br>(-1.27)          | -0.868**<br>(-2.52)  | -0.235<br>(-1.27)          | -0.024<br>(-0.11)          | -1.038***<br>(-3.02) | -0.129<br>(-0.71)          |
| <i>Log Price</i>                              | -1.063<br>(-1.13)          | -1.047<br>(-1.12)    | -0.482<br>(-1.05)          | -0.671<br>(-0.71)          | -0.735<br>(-0.83)    | -0.029<br>(-0.07)          |
| <i>Return Volatility</i>                      | -59.587***<br>(-3.08)      | 27.445*<br>(1.66)    | 14.899<br>(1.34)           | -53.652***<br>(-2.87)      | 19.210<br>(1.20)     | 11.467<br>(1.05)           |
| <i>Analyst Dispersion</i>                     | -23.636<br>(-0.40)         | -40.842<br>(-0.71)   | -15.946<br>(-0.40)         | -11.993<br>(-0.20)         | -17.323<br>(-0.30)   | -2.625<br>(-0.07)          |
| <i>Busyness</i>                               | -0.104<br>(-0.73)          | -0.245*<br>(-1.83)   | -0.172*<br>(-1.79)         | -0.132<br>(-0.93)          | -0.219*<br>(-1.66)   | -0.164*<br>(-1.72)         |
| $ \Delta IB $                                 | 16.729***<br>(2.98)        | -4.012<br>(-0.78)    | 2.450<br>(0.69)            | 17.171***<br>(3.09)        | -4.609<br>(-0.91)    | 2.027<br>(0.58)            |
| $ \Delta Shares $                             | 3.721<br>(0.79)            | -1.964<br>(-0.51)    | 1.505<br>(0.52)            | 3.251<br>(0.72)            | -2.171<br>(-0.57)    | 1.376<br>(0.48)            |
| $ SpecItems $                                 | -18.458<br>(-1.50)         | -8.991<br>(-0.79)    | -10.912<br>(-1.33)         | -21.400*<br>(-1.77)        | -7.112<br>(-0.64)    | -9.847<br>(-1.22)          |
| <i>SBC High</i>                               | -0.149<br>(-0.57)          | 0.174<br>(0.77)      | 0.106<br>(0.65)            | -0.168<br>(-0.66)          | 0.186<br>(0.83)      | 0.095<br>(0.59)            |
| <i>Unexpected Item</i>                        | -0.258<br>(-0.97)          | -0.036<br>(-0.17)    | -0.123<br>(-0.78)          | -0.222<br>(-0.89)          | -0.023<br>(-0.11)    | -0.103<br>(-0.68)          |
| <i>Stock Split</i>                            | -2.679**<br>(-2.21)        | -1.110<br>(-0.92)    | -1.584*<br>(-1.90)         | -2.266*<br>(-1.78)         | -0.472<br>(-0.42)    | -0.977<br>(-1.18)          |

|                                |                    |                     |                   |                    |                   |                    |
|--------------------------------|--------------------|---------------------|-------------------|--------------------|-------------------|--------------------|
| <i>Fiscal Q4</i>               | 0.600***<br>(2.83) | -0.045<br>(-0.24)   | 0.276**<br>(2.07) | 0.606***<br>(2.89) | 0.230<br>(1.23)   | 0.417***<br>(3.19) |
| <i>Institutional Ownership</i> | 3.234<br>(1.56)    | -0.702**<br>(-2.02) | -0.470<br>(-1.41) | 1.626<br>(0.83)    | -0.543<br>(-1.57) | -0.416<br>(-1.25)  |
| <i>Log # Analysts</i>          | 0.508<br>(0.79)    | -0.218<br>(-0.39)   | -0.248<br>(-0.65) | 0.648<br>(1.04)    | -0.328<br>(-0.60) | -0.371<br>(-0.99)  |
| Controls                       | Yes                | Yes                 | Yes               | Yes                | Yes               | Yes                |
| Firm FE                        | Yes                | Yes                 | Yes               | Yes                | Yes               | Yes                |
| Year FE                        | Yes                | Yes                 | Yes               | Yes                | Yes               | Yes                |
| Observations                   | 9,405              | 14,402              | 23,833            | 9,405              | 14,402            | 23,833             |
| Adjusted $R^2$                 | 0.085              | 0.106               | 0.080             | 0.092              | 0.110             | 0.083              |

**Panel B. Horse race:  $SUE^{Acc}$  vs  $SUE^{IBES}$ .**

| Dep. Var.:                                    | Mkt $CAR_{[-1,+1]}$ |                     |                            | FFC $CAR_{[-1,+1]}$ |                     |                            |
|-----------------------------------------------|---------------------|---------------------|----------------------------|---------------------|---------------------|----------------------------|
|                                               | (1)<br>Pre          | (2)<br>Post         | (3)<br>Pooled              | (4)<br>Pre          | (5)<br>Post         | (6)<br>Pooled              |
| $SUE^{Acc} (\beta_1)$                         | 1.428***<br>(4.00)  | 3.678***<br>(10.49) | 1.430***<br>(4.31)         | 1.527***<br>(4.36)  | 3.796***<br>(10.70) | 1.494***<br>(4.48)         |
| $SUE^{Acc} \times Post (\beta_3)$             |                     |                     | 2.153***<br>(4.82)         |                     |                     | 2.204***<br>(4.98)         |
| $SUE^{IBES} (\beta_2)$                        | 2.176***<br>(7.08)  | 1.460***<br>(4.44)  | 2.054***<br>(7.31)         | 2.149***<br>(7.20)  | 1.407***<br>(4.20)  | 1.974***<br>(7.01)         |
| $SUE^{IBES} \times Post (\beta_4)$            |                     |                     | -0.519<br>(-1.34)          |                     |                     | -0.484<br>(-1.28)          |
| Premium (Pre-ChatGPT)                         | -0.747<br>(-1.20)   |                     | -0.623<br>(-1.09)          | -0.622<br>(-1.03)   |                     | -0.480<br>(-0.83)          |
| Premium (Post-ChatGPT)                        |                     | 2.219***<br>(3.56)  | 2.049***<br>(3.60)         |                     | 2.389***<br>(3.75)  | 2.208***<br>(3.82)         |
| <b><math>\Delta</math> Premium (Post-Pre)</b> |                     |                     | <b>2.672***<br/>(3.44)</b> |                     |                     | <b>2.688***<br/>(3.50)</b> |
| Controls                                      | Yes                 | Yes                 | Yes                        | Yes                 | Yes                 | Yes                        |
| Firm FE                                       | Yes                 | Yes                 | Yes                        | Yes                 | Yes                 | Yes                        |
| Year FE                                       | Yes                 | Yes                 | Yes                        | Yes                 | Yes                 | Yes                        |
| Observations                                  | 9,405               | 14,402              | 23,833                     | 9,405               | 14,402              | 23,833                     |
| Adjusted $R^2$                                | 0.087               | 0.106               | 0.080                      | 0.094               | 0.110               | 0.083                      |

*Notes.* This table reports OLS estimates of the horse race between the accuracy-weighted surprise  $SUE^{Acc}$  and a benchmark surprise  $SUE^X$ ,  $CAR_{i,q} = \alpha + \beta_1 SUE_{i,q}^{Acc} + \beta_2 SUE_{i,q}^X + \beta_3 SUE_{i,q}^{Acc} \times Post_q + \beta_4 SUE_{i,q}^X \times Post_q + \text{controls} + \text{FE} + \varepsilon_{i,q}$ , where  $Post_q$  equals one for announcements on or after November 30, 2022 (the ChatGPT release).  $SUE^{Acc}$  uses softmax weights based on each provider's within-firm-quarter z-scored four-quarter rolling mean absolute forecast error (MAFE). **Panel A** uses the equal-weighted benchmark  $SUE^{EW5}$  and **Panel B** the I/B/E/S benchmark  $SUE^{IBES}$ ; both panels use the same firm and event controls, which Panel B suppresses for brevity. Columns (1) and (4) restrict the sample to pre-ChatGPT announcements and columns (2) and (5) to post-ChatGPT announcements, so the  $Post$  interactions drop out; columns (3) and (6) pool both periods. Columns (1)–(3) use the market-adjusted  $CAR_{[-1,+1]}$ ; columns (4)–(6) the Fama-French-Carhart (FFC) four-factor  $CAR_{[-1,+1]}$ . The *Premium* rows (Pre-ChatGPT, Post-ChatGPT) report the accuracy premium, the loading on  $SUE^{Acc}$  minus the loading on the benchmark:  $\beta_1 - \beta_2$  in the single-period columns and in the pre-ChatGPT row of the pooled columns, and  $(\beta_1 + \beta_3) - (\beta_2 + \beta_4)$  in the

post-ChatGPT row of the pooled columns; the bolded  $\Delta$  Premium row reports the post-pre change,  $\beta_3 - \beta_4$ . All columns include firm and year fixed effects and the full control set; standard errors are clustered by firm; continuous variables are winsorized at (1, 99).  $t$ -statistics in parentheses; \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ . Variable definitions are in [Appendix A](#).

**TABLE 8. The Accuracy Premium Across GPT-Capability Regimes**

**Panel A. Benchmark = Equal-Weighted Five-FDP Consensus**

| Regime:                                         | (1) R0<br>Pre-ChatGPT           | (2) R1<br>text-only           | (3) R2<br>browsing              | (4) R3<br>native search         |
|-------------------------------------------------|---------------------------------|-------------------------------|---------------------------------|---------------------------------|
| $SUE^{Acc} (\beta_1)$                           | 1.376***<br>(3.45)              | 2.597***<br>(3.38)            | 3.753***<br>(6.19)              | 3.954***<br>(5.69)              |
| Benchmark $SUE (\beta_2)$                       | 2.324***<br>(6.10)              | 2.419***<br>(3.78)            | 1.325***<br>(2.63)              | 1.404**<br>(2.26)               |
| <b>Premium (<math>\beta_1 - \beta_2</math>)</b> | <b>-0.948</b><br><b>(-1.28)</b> | <b>0.178</b><br><b>(0.13)</b> | <b>2.429**</b><br><b>(2.35)</b> | <b>2.550**</b><br><b>(2.07)</b> |
| Controls / Firm FE / Year FE                    | Yes                             | Yes                           | Yes                             | Yes                             |
| Observations                                    | 9,405                           | 3,705                         | 5,529                           | 5,084                           |
| Adjusted $R^2$                                  | 0.085                           | 0.158                         | 0.151                           | 0.140                           |

**Panel B. Benchmark = I/B/E/S Consensus**

| Regime:                                         | (1) R0<br>Pre-ChatGPT           | (2) R1<br>text-only             | (3) R2<br>browsing            | (4) R3<br>native search          |
|-------------------------------------------------|---------------------------------|---------------------------------|-------------------------------|----------------------------------|
| $SUE^{Acc} (\beta_1)$                           | 1.428***<br>(4.00)              | 1.918**<br>(2.35)               | 3.362***<br>(6.17)            | 5.079***<br>(7.87)               |
| Benchmark $SUE (\beta_2)$                       | 2.176***<br>(7.08)              | 3.144***<br>(4.70)              | 1.842***<br>(3.74)            | 0.169<br>(0.30)                  |
| <b>Premium (<math>\beta_1 - \beta_2</math>)</b> | <b>-0.747</b><br><b>(-1.20)</b> | <b>-1.226</b><br><b>(-0.86)</b> | <b>1.519</b><br><b>(1.61)</b> | <b>4.909***</b><br><b>(4.41)</b> |
| Controls / Firm FE / Year FE                    | Yes                             | Yes                             | Yes                           | Yes                              |
| Observations                                    | 9,405                           | 3,705                           | 5,529                         | 5,084                            |
| Adjusted $R^2$                                  | 0.087                           | 0.170                           | 0.155                         | 0.138                            |

*Notes.* This table estimates the accuracy premium separately within each GPT-capability regime. For each regime the specification is  $CAR_{[-1,+1]}^{Mkt} = \alpha + \beta_1 SUE_{i,q}^{Acc} + \beta_2 SUE_{i,q}^X + \text{controls} + \alpha_i + \tau_t + \varepsilon$ , and the *Accuracy Premium* is the contrast  $\beta_1 - \beta_2$ . Regimes are defined by announcement date: **R0** (before 2022-11-30, Pre-ChatGPT), **R1** (2022-11-30 to 2023-09-26, text-only), **R2** (2023-09-27 to 2024-10-30, browsing), and **R3** (from 2024-10-31, native search). The benchmark  $X$  is the equal-weighted five-FDP consensus (Panel A) or I/B/E/S (Panel B). All columns include the firm and event controls listed in Table 7, firm ( $\alpha_i$ ) and year ( $\tau_t$ ) fixed effects, and firm-clustered standard errors; continuous variables are winsorized at (1, 99).  $t$ -statistics in parentheses; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Variable definitions are in [Appendix A](#).

**TABLE 9. Cross-Sectional Tests of the Accuracy Premium**

**Panel A. Concurrent Cross-FDP SUE Dispersion.**

| Benchmark:                                    | Equal-Weighted                |                                 | IBES                            |                                  |
|-----------------------------------------------|-------------------------------|---------------------------------|---------------------------------|----------------------------------|
|                                               | (1) Low                       | (2) High                        | (3) Low                         | (4) High                         |
| $SUE^{Acc} (\beta_1)$                         | 4.138***<br>(6.37)            | 1.433***<br>(4.23)              | 1.569**<br>(2.43)               | 1.748***<br>(5.57)               |
| Benchmark $SUE (\beta_2)$                     | 3.318***<br>(5.86)            | 2.024***<br>(6.41)              | 7.496***<br>(10.03)             | 1.560***<br>(6.10)               |
| $SUE^{Acc} \times Post^{Scr} (\beta_3)$       | 2.748**<br>(2.48)             | 1.610***<br>(2.89)              | 1.439<br>(1.34)                 | 1.746***<br>(3.63)               |
| Benchmark $SUE \times Post^{Scr} (\beta_4)$   | 0.712<br>(0.76)               | -0.602<br>(-1.20)               | 2.130*<br>(1.86)                | -0.681*<br>(-1.70)               |
| Premium (Pre-Scrape)                          | 0.821<br>(0.73)               | -0.591<br>(-0.96)               | -5.927***<br>(-4.58)            | 0.188<br>(0.35)                  |
| Premium (Post-Scrape)                         | 2.857*<br>(1.71)              | 1.620**<br>(1.98)               | -6.619***<br>(-3.81)            | 2.615***<br>(3.81)               |
| <b><math>\Delta</math> Premium (Post-Pre)</b> | <b>2.037</b><br><b>(1.06)</b> | <b>2.211**</b><br><b>(2.21)</b> | <b>-0.691</b><br><b>(-0.33)</b> | <b>2.427***</b><br><b>(2.99)</b> |
| Controls                                      | Yes                           | Yes                             | Yes                             | Yes                              |
| Firm FE                                       | Yes                           | Yes                             | Yes                             | Yes                              |
| Year FE                                       | Yes                           | Yes                             | Yes                             | Yes                              |
| Observations                                  | 11,832                        | 11,846                          | 11,832                          | 11,846                           |
| Adjusted $R^2$                                | 0.111                         | 0.075                           | 0.127                           | 0.075                            |

**Panel B. Special-Items Magnitude.**

| Benchmark:                                    | Equal-Weighted                  |                                 | IBES                            |                                  |
|-----------------------------------------------|---------------------------------|---------------------------------|---------------------------------|----------------------------------|
|                                               | (1) Low                         | (2) High                        | (3) Low                         | (4) High                         |
| $SUE^{Acc} (\beta_1)$                         | 2.345***<br>(4.36)              | 1.613***<br>(4.49)              | 1.940***<br>(4.34)              | 1.969***<br>(5.48)               |
| Benchmark $SUE (\beta_2)$                     | 2.153***<br>(4.21)              | 2.078***<br>(6.41)              | 2.465***<br>(6.46)              | 1.615***<br>(5.25)               |
| $SUE^{Acc} \times Post^{Scr} (\beta_3)$       | 0.247<br>(0.31)                 | 2.035***<br>(3.39)              | 0.440<br>(0.58)                 | 2.583***<br>(4.69)               |
| Benchmark $SUE \times Post^{Scr} (\beta_4)$   | 0.799<br>(1.14)                 | -0.483<br>(-0.92)               | 0.659<br>(0.98)                 | -1.055**<br>(-2.28)              |
| Premium (Pre-Scrape)                          | 0.192<br>(0.19)                 | -0.464<br>(-0.74)               | -0.525<br>(-0.68)               | 0.354<br>(0.58)                  |
| Premium (Post-Scrape)                         | -0.360<br>(-0.28)               | 2.054**<br>(2.48)               | -0.743<br>(-0.65)               | 3.992***<br>(5.36)               |
| <b><math>\Delta</math> Premium (Post-Pre)</b> | <b>-0.552</b><br><b>(-0.39)</b> | <b>2.518**</b><br><b>(2.40)</b> | <b>-0.219</b><br><b>(-0.16)</b> | <b>3.638***</b><br><b>(3.92)</b> |
| Controls                                      | Yes                             | Yes                             | Yes                             | Yes                              |
| Firm FE                                       | Yes                             | Yes                             | Yes                             | Yes                              |
| Year FE                                       | Yes                             | Yes                             | Yes                             | Yes                              |
| Observations                                  | 11,823                          | 11,828                          | 11,823                          | 11,828                           |
| Adjusted $R^2$                                | 0.089                           | 0.073                           | 0.095                           | 0.071                            |

*Notes.* This table estimates the accuracy-weighted horse race against the equal-weighted (EW5) and I/B/E/S

benchmarks separately for two cross-sectional partitions. **Panel A** splits firm-quarters at the median of the interquartile range of the five providers' SUE for the same firm-quarter. **Panel B** splits firm-quarters at the median of  $|SpecItems|$  (%), absolute special items scaled by absolute income before extraordinary items ([Appendix A](#), Panel C), not the asset-scaled control of the same name. Within each panel, columns (1)–(2) use  $SUE^{EW5}$  and columns (3)–(4) use  $SUE^{IBES}$ . The Accuracy Premium is the difference between the loading on  $SUE^{Acc}$  and the loading on the benchmark SUE ( $\beta_1 - \beta_2$  pre-scrape,  $(\beta_1 + \beta_3) - (\beta_2 + \beta_4)$  post-scrape); the bolded  $\Delta$  row reports the post-pre change,  $\beta_3 - \beta_4$ . Here  $Post^{Scr}$  (post-scraping) equals one for announcements on or after the onset of GPT browsing (2023-09-27), distinct from the post-ChatGPT  $Post$  used in the other tables. All panels include the full control vector, firm fixed effects, year fixed effects, and firm-clustered standard errors. The dependent variable is the market-adjusted  $CAR_{[-1,+1]}$ . Continuous variables are winsorized at the (1, 99) level.  $t$ -statistics in parentheses; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Each panel's sample comprises the firm-quarters with non-missing values for all regression variables. Variable definitions are in [Appendix A](#).

**TABLE 10. LLM Visibility Is Distinct from Classical Prominence Proxies****Panel A. Overlap of high-visibility classifications (%)**

|                       | LLM<br>Visibility | Firm<br>Size | Analyst<br>Coverage | Inst<br>Ownership |
|-----------------------|-------------------|--------------|---------------------|-------------------|
| <b>LLM Visibility</b> | 100.0             | 50.2         | 49.7                | 47.1              |
| Firm Size             | 51.0              | 100.0        | 79.6                | 40.7              |
| Analyst Coverage      | 50.6              | 79.8         | 100.0               | 44.5              |
| Inst Ownership        | 47.9              | 40.7         | 44.5                | 100.0             |

**Panel B. Accuracy-premium concentration: Equal-weighted benchmark**

| Visibility measure           | Univariate $\Delta$ Premium           |                                    | Horse-race              |
|------------------------------|---------------------------------------|------------------------------------|-------------------------|
|                              | High group                            | Low group                          | triple diff.            |
| <b>LLM Visibility</b>        | <b>3.263***</b><br>(2.59)<br>[11,219] | <b>0.300</b><br>(0.25)<br>[11,825] | <b>3.172*</b><br>(1.80) |
| Firm Size                    | 0.723<br>(0.55)<br>[11,901]           | 2.447**<br>(2.18)<br>[11,143]      | 0.111<br>(0.05)         |
| Analyst Coverage             | 0.109<br>(0.08)<br>[11,861]           | 3.076**<br>(2.57)<br>[11,183]      | -3.161<br>(-1.38)       |
| Inst Ownership               | 3.224**<br>(2.51)<br>[11,593]         | 0.646<br>(0.58)<br>[11,451]        | 3.222*<br>(1.86)        |
| Controls / Firm FE / Year FE | Yes                                   | Yes                                | Yes                     |
| Observations                 | 23,044                                |                                    | 23,044                  |

**Panel C. Accuracy-premium concentration: I/B/E/S benchmark**

| Visibility measure           | Univariate $\Delta$ Premium           |                                      | Horse-race                |
|------------------------------|---------------------------------------|--------------------------------------|---------------------------|
|                              | High group                            | Low group                            | triple diff.              |
| <b>LLM Visibility</b>        | <b>5.743***</b><br>(5.67)<br>[11,219] | <b>-0.125</b><br>(-0.12)<br>[11,825] | <b>5.841***</b><br>(4.03) |
| Firm Size                    | 2.191<br>(1.28)<br>[11,901]           | 2.462***<br>(2.76)<br>[11,143]       | 2.010<br>(0.82)           |
| Analyst Coverage             | 0.441<br>(0.29)<br>[11,861]           | 3.198***<br>(3.59)<br>[11,183]       | -4.099*<br>(-1.77)        |
| Inst Ownership               | 2.564**<br>(2.46)<br>[11,593]         | 2.418**<br>(2.08)<br>[11,451]        | 0.625<br>(0.41)           |
| Controls / Firm FE / Year FE | Yes                                   | Yes                                  | Yes                       |
| Observations                 | 23,044                                |                                      | 23,044                    |

Notes. Each measure—LLM visibility, firm size (log market equity), analyst coverage (log average analyst

count), and institutional ownership—is split at its cross-firm median; the visibility score is discrete, so firms tied at the median join the Low group, which is why the split is not exactly even. **Panel A:** cell  $(i, j)$  is the percentage of firms in measure  $i$ 's High group that measure  $j$  also classifies as High (diagonal = 100). **Panels B and C:**  $\Delta\text{Premium}$  is the post-pre change in the accuracy premium  $\beta(SUE^{Acc}) - \beta(SUE^X)$  against benchmark  $X$  = the equal-weighted five-FDP consensus (Panel B) or I/B/E/S (Panel C), estimated within each measure's High and Low group; the horse-race triple  $(SUE^{Acc} - SUE^X) \times \text{Post}^{Scr} \times \text{High}_m$  is from one pooled regression including the triple for every measure jointly. Throughout,  $\text{Post}^{Scr}$  is the post-scraping indicator (announcements on or after the onset of browsing, 2023-09-27), as in the cross-sectional tests (Table 9). The dependent variable throughout Panels B and C is the market-adjusted  $CAR_{[-1,+1]}$ . All specifications include the firm and event controls listed in Table 7, firm and year fixed effects, and firm-clustered standard errors; continuous variables are winsorized at (1, 99).  $t$ -statistics in parentheses and group observation counts in brackets (each measure's High and Low groups partition the same 23,044 firm-quarters, the announcements with non-missing values for all regression variables); \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Variable definitions are in Appendix A.



**TABLE 11. LLM Visibility and the Accuracy Premium by GPT-Capability Regime**

**Panel A. Equal-Weighted Five-FDP Benchmark.**

| Regime:                                                                                              | (1) R0<br>Pre-ChatGPT | (2) R1<br>text-only | (3) R2<br>browsing               | (4) R3<br>native search          |
|------------------------------------------------------------------------------------------------------|-----------------------|---------------------|----------------------------------|----------------------------------|
| $SUE^{Acc} (\beta_1)$                                                                                | 1.361**<br>(2.57)     | 3.503***<br>(5.05)  | 3.077***<br>(2.95)               | 3.055***<br>(4.18)               |
| $SUE^{EW5} (\beta_2)$                                                                                | 2.252***<br>(4.09)    | 2.185***<br>(3.61)  | 1.885***<br>(3.65)               | 2.002***<br>(3.21)               |
| $SUE^{Acc} \times High\ Visibility (\beta_3)$                                                        | 0.020<br>(0.05)       | -2.274<br>(-1.49)   | 1.550*<br>(1.79)                 | 1.815*<br>(1.93)                 |
| $SUE^{EW5} \times High\ Visibility (\beta_4)$                                                        | 0.153<br>(0.28)       | 0.889<br>(0.79)     | -1.133**<br>(-2.11)              | -1.269<br>(-1.51)                |
| <i>Accuracy premium = loading on <math>SUE^{Acc}</math> - loading on benchmark <math>SUE</math>:</i> |                       |                     |                                  |                                  |
| Premium (Low Vis)                                                                                    | -0.892<br>(-0.90)     | 1.317<br>(1.12)     | 1.192<br>(0.89)                  | 1.053<br>(0.83)                  |
| Premium (High Vis)                                                                                   | -1.025<br>(-1.57)     | -1.845<br>(-0.94)   | <b>3.875***</b><br><b>(4.60)</b> | <b>4.136***</b><br><b>(4.16)</b> |
| <b>Premium (High - Low)</b>                                                                          | -0.133<br>(-0.15)     | -3.163<br>(-1.22)   | <b>2.683**</b><br><b>(2.12)</b>  | <b>3.083*</b><br><b>(1.85)</b>   |
| Controls / Firm FE / Year-qtr FE                                                                     | Yes                   | Yes                 | Yes                              | Yes                              |
| Observations                                                                                         | 9,399                 | 3,705               | 5,529                            | 5,084                            |
| Adjusted $R^2$                                                                                       | 0.086                 | 0.163               | 0.165                            | 0.143                            |

**Panel B. I/B/E/S Benchmark.**

| Regime:                                                                                              | (1) R0<br>Pre-ChatGPT | (2) R1<br>text-only | (3) R2<br>browsing               | (4) R3<br>native search           |
|------------------------------------------------------------------------------------------------------|-----------------------|---------------------|----------------------------------|-----------------------------------|
| $SUE^{Acc} (\beta_1)$                                                                                | 1.607***<br>(4.07)    | 3.138***<br>(3.15)  | 2.737***<br>(6.05)               | 4.029***<br>(8.81)                |
| $SUE^{IBES} (\beta_2)$                                                                               | 1.859***<br>(3.60)    | 2.424***<br>(2.96)  | 2.403**<br>(2.28)                | 0.974**<br>(2.13)                 |
| $SUE^{Acc} \times High\ Visibility (\beta_3)$                                                        | -0.382<br>(-1.03)     | -2.353*<br>(-1.74)  | 1.667**<br>(2.44)                | 2.376***<br>(3.68)                |
| $SUE^{IBES} \times High\ Visibility (\beta_4)$                                                       | 0.669<br>(0.97)       | 1.362<br>(1.45)     | -1.380<br>(-1.45)                | -1.856***<br>(-3.06)              |
| <i>Accuracy premium = loading on <math>SUE^{Acc}</math> - loading on benchmark <math>SUE</math>:</i> |                       |                     |                                  |                                   |
| Premium (Low Vis)                                                                                    | -0.252<br>(-0.31)     | 0.714<br>(0.41)     | 0.335<br>(0.26)                  | 3.054***<br>(4.00)                |
| Premium (High Vis)                                                                                   | -1.303**<br>(-2.06)   | -3.001**<br>(-2.12) | <b>3.382***</b><br><b>(6.17)</b> | <b>7.286***</b><br><b>(10.56)</b> |
| <b>Premium (High - Low)</b>                                                                          | -1.052<br>(-1.07)     | -3.715*<br>(-1.72)  | <b>3.047**</b><br><b>(2.05)</b>  | <b>4.232***</b><br><b>(3.93)</b>  |
| Controls / Firm FE / Year-qtr FE                                                                     | Yes                   | Yes                 | Yes                              | Yes                               |
| Observations                                                                                         | 9,399                 | 3,705               | 5,529                            | 5,084                             |
| Adjusted $R^2$                                                                                       | 0.088                 | 0.175               | 0.169                            | 0.142                             |

*Notes.* This table estimates the accuracy-weighted horse race  $CAR_{[-1,+1]}^{Mkt} = \alpha + \beta_1 SUE_{i,q}^{Acc} + \beta_2 SUE_{i,q}^X + \beta_3 SUE_{i,q}^{Acc} \times High\ Visibility_i + \beta_4 SUE_{i,q}^X \times High\ Visibility_i + \text{controls} + FE + \varepsilon$  separately within each GPT-capability regime, where  $SUE_{i,q}^{Acc}$  is the accuracy-weighted multi-FDP surprise with softmax weights based on each provider's within-firm-quarter z-scored four-quarter rolling MAFE. *High Visibility<sub>i</sub>* is an indicator equal to one for firms with above-median LLM visibility. Regimes track ChatGPT's information-retrieval capability by announcement date: **R0** (before 2022-11-30, Pre-ChatGPT), **R1** (2022-11-30 to 2023-09-26, text-only), **R2** (2023-09-27 to 2024-10-30, browsing / Bing), and **R3** (from 2024-10-31, native ChatGPT Search). **Panel A** uses the equal-weighted five-FDP benchmark  $SUE^{EW5}$ ; **Panel B** the I/B/E/S benchmark. The *Accuracy Premium* is the contrast  $SUE^{Acc}$ -Benchmark SUE, reported for low- and high-visibility firms (Premium (Low Vis) =  $\beta_1 - \beta_2$ ; Premium (High Vis) =  $(\beta_1 + \beta_3) - (\beta_2 + \beta_4)$ ); the bolded Premium (High - Low) row is the triple difference  $\beta_3 - \beta_4$ . All specifications include the full control vector, firm fixed effects, and year-quarter fixed effects; the dependent variable is the market-adjusted  $CAR_{[-1,+1]}$ . Each regime sample comprises the announcements with non-missing values for all regression variables. Continuous variables are winsorized at the (1, 99) level; standard errors are two-way clustered by firm and calendar year. *t*-statistics in parentheses; \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Variable definitions are in [Appendix A](#).

**TABLE 12. ChatGPT Outages and the Accuracy Premium**

| Dep. Var.: day-0 AR <sub>0</sub> (market-adj., %) | Benchmark:<br>Equal-Weighted<br>(1) | Benchmark:<br>IBES<br>(2)                    |
|---------------------------------------------------|-------------------------------------|----------------------------------------------|
| $SUE^{Acc} (\beta_1)$                             | 1.631 <sup>***</sup><br>(6.93)      | 1.599 <sup>***</sup><br>(7.78)               |
| $SUE^{Acc} \times Outage (\beta_3)$               | 1.178<br>(0.80)                     | -1.685 <sup>*</sup><br>(-1.85)               |
| Benchmark $SUE (\beta_2)$                         | 0.626 <sup>***</sup><br>(3.13)      | 0.654 <sup>***</sup><br>(4.10)               |
| Benchmark $SUE \times Outage (\beta_4)$           | -2.608 <sup>*</sup><br>(-1.79)      | 1.488 <sup>**</sup><br>(2.27)                |
| Premium (non-outage)                              | 1.005 <sup>**</sup><br>(2.48)       | 0.945 <sup>***</sup><br>(2.88)               |
| Premium (outage)                                  | 4.792 <sup>*</sup><br>(1.77)        | -2.228 <sup>*</sup><br>(-1.65)               |
| <b><math>\Delta</math> Premium (Outage-Non)</b>   | <b>3.787</b><br><b>(1.37)</b>       | <b>-3.173<sup>**</sup></b><br><b>(-2.29)</b> |
| Controls / Firm FE / Year FE                      | Yes                                 | Yes                                          |
| Observations                                      | 14,402                              | 14,402                                       |
| Adjusted $R^2$                                    | 0.051                               | 0.052                                        |

*Notes.* This table tests whether the accuracy premium collapses on earnings announcements that follow a major ChatGPT outage. The sample is post-ChatGPT firm-quarters;  $Outage_{i,q} = 1$  if a ChatGPT incident of at least 240 minutes (from the OpenAI incident log) occurred on the calendar day immediately preceding the announcement (84 outage firm-quarters). The dependent variable is the day-0 market-adjusted abnormal return; both columns include firm and year fixed effects, the full control set, and firm-clustered standard errors. The Accuracy Premium is the loading on  $SUE^{Acc}$  minus the loading on the benchmark SUE; the bolded row reports its outage-non-outage difference.  $t$ -statistics in parentheses; <sup>\*</sup> $p < 0.10$ , <sup>\*\*</sup> $p < 0.05$ , <sup>\*\*\*</sup> $p < 0.01$ . Variable definitions are in [Appendix A](#).

## Appendix A: Variable Definitions

| Variable                                                                                                                         | Definition                                                                                                                                                                                                                 | Source                                        |
|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Panel A. Across-FDP firm-quarter outcome and surprise variables (indexed by firm <math>i</math>, quarter <math>q</math>).</b> |                                                                                                                                                                                                                            |                                               |
| $CAR_{i,q,[-1,+1]}^{Mkt}$                                                                                                        | Three-day market-adjusted cumulative abnormal return around the earnings-announcement date, $\sum_{t=-1}^{+1} (r_{i,t} - vwret_d_t)$ , in percent.                                                                         | CRSP                                          |
| $CAR_{i,q,[-1,+1]}^{FFC}$                                                                                                        | Three-day FFC four-factor (FF3+UMD) abnormal CAR over $[-1, +1]$ with factor loadings estimated on the $[-260, -11]$ window.                                                                                               | CRSP                                          |
| $BHAR_{i,q,[-2,+1]}^{Mkt}$                                                                                                       | Four-day market-adjusted buy-and-hold abnormal return over $[-2, +1]$ .                                                                                                                                                    | CRSP                                          |
| $BHAR_{i,q,[-2,+1]}^{FFC}$                                                                                                       | Four-day FFC four-factor buy-and-hold abnormal return over $[-2, +1]$ .                                                                                                                                                    | CRSP                                          |
| $SUE_{i,q}^g$                                                                                                                    | Per-FDP standardized unexpected earnings for provider $g \in \{\text{FactSet, Bloomberg, Zacks, CapIQ, IBES}\}$ : $(\text{actual EPS}_{i,q}^g - \text{consensus EPS}_{i,q}^g) / \text{price}_{i,q}$ , in percent of price. | FactSet; Bloomberg; Zacks; S&P CapIQ; I/B/E/S |
| $SUE_{i,q}^{Acc}$                                                                                                                | Accuracy-weighted multi-vendor SUE, $\sum_g w_{i,q}^g SUE_{i,q}^g$ , where $w_{i,q}^g$ are softmax weights based on each FDP's $z$ -scored four-quarter rolling MAFE.                                                      | Calculated                                    |
| $SUE_{i,q}^{EW5}$                                                                                                                | Equal-weighted multi-vendor SUE, $(1/5) \sum_g SUE_{i,q}^g$ .                                                                                                                                                              | Calculated                                    |
| $Standardized Pred. Surp_{i,q}^X$                                                                                                | Min-max standardized predicted surprise $(Consensus_{i,q}^{Acc} - Consensus_{i,q}^X) / \text{price}_{i,q}$ for benchmark $X \in \{EW5, IBES\}$ .                                                                           | Calculated                                    |
| <b>Panel B. Firm and event controls (indexed by firm <math>i</math>, quarter <math>q</math>).</b>                                |                                                                                                                                                                                                                            |                                               |
| $Log Market Equity_{i,q-1}$                                                                                                      | Natural log of firm $i$ 's market capitalization at the end of quarter $q-1$ .                                                                                                                                             | CRSP                                          |
| $Book-to-Market_{i,q-1}$                                                                                                         | Book value of equity divided by market value of equity at quarter-end $q-1$ .                                                                                                                                              | Compustat; CRSP                               |
| $Leverage_{i,q-1}$                                                                                                               | Long-term debt plus debt in current liabilities, scaled by total assets, at $q-1$ .                                                                                                                                        | Compustat                                     |
| $Prior 12-Month Return_{i,q}$                                                                                                    | Buy-and-hold raw return over the twelve months ending one month before firm $i$ 's quarter- $q$ announcement.                                                                                                              | CRSP                                          |
| $Log Price_{i,q-1}$                                                                                                              | Natural log of share price at quarter-end $q-1$ .                                                                                                                                                                          | CRSP                                          |
| $Return Volatility_{i,q}$                                                                                                        | Standard deviation of daily returns over the $[-260, -11]$ window before the announcement.                                                                                                                                 | CRSP                                          |
| $Analyst Dispersion_{i,q}$                                                                                                       | Standard deviation of analyst EPS forecasts divided by absolute consensus.                                                                                                                                                 | I/B/E/S                                       |
| $Busyness_{i,q}$                                                                                                                 | Natural log of the number of S&P 1500 firms announcing earnings on firm $i$ 's announcement day.                                                                                                                           | Compustat                                     |

(continued on next page)

| Variable                                                                   | Definition                                                                                                                                                                                                                                                                                                                           | Source              |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| $ \Delta IB _{i,q}$                                                        | Absolute year-over-year change in income before extraordinary items, scaled by lagged total assets.                                                                                                                                                                                                                                  | Compustat           |
| $ \Delta Shares _{i,q}$                                                    | Absolute year-over-year change in shares outstanding, scaled by lagged shares.                                                                                                                                                                                                                                                       | Compustat           |
| $ SpecItems _{i,q}$                                                        | Absolute value of special items, scaled by lagged total assets.                                                                                                                                                                                                                                                                      | Compustat           |
| $SBC\ High_{i,q}$                                                          | Indicator equal to one if stock-based compensation expense exceeds the sample median.                                                                                                                                                                                                                                                | Compustat           |
| $Unexpected\ Item_{i,q}$                                                   | Indicator equal to one if the quarter contains a non-recurring item flagged by Compustat.                                                                                                                                                                                                                                            | Compustat           |
| $Stock\ Split_{i,q}$                                                       | Indicator equal to one if a stock split was effective within [-30, +30] trading days of the announcement.                                                                                                                                                                                                                            | CRSP                |
| $Fiscal\ Q4_{i,q}$                                                         | Indicator equal to one for fiscal fourth-quarter announcements.                                                                                                                                                                                                                                                                      | Compustat           |
| $Institutional\ Ownership_{i,q-1}$                                         | Fraction of shares held by institutional investors at $q-1$ .                                                                                                                                                                                                                                                                        | Thomson Reuters 13F |
| $\#Analysts_{i,q}$                                                         | Number of unique analysts in the IBES detail file forecasting EPS for firm-quarter $(i, q)$ .                                                                                                                                                                                                                                        | I/B/E/S             |
| <b>Panel C. Cross-sectional partitioning and identification variables.</b> |                                                                                                                                                                                                                                                                                                                                      |                     |
| $LLM\ Visibility_i$                                                        | EPS-extraction indicator $\times \log(1 + \text{unique cited domains})$ , averaged across ChatGPT and Perplexity for direct queries about firm $i$ 's quarterly EPS.                                                                                                                                                                 | DataForSEO          |
| $Cross-FDP\ SUE\ Dispersion_{i,q}$                                         | Inter-quartile range of $SUE_{i,q}^g$ across the five FDPs in firm-quarter $(i, q)$ .                                                                                                                                                                                                                                                | Calculated          |
| $ SpecItems _{i,q} (\%)$                                                   | Absolute special items as a percentage of absolute income before extraordinary items, $ SpecItems _{i,q}/ IB _{i,q} \times 100$ .                                                                                                                                                                                                    | Compustat           |
| $IBES\ Up_i$                                                               | Indicator equal to one if IBES's rounded average per-firm-quarter accuracy rank among the five FDPs rose by at least two positions from pre- to post-ChatGPT.                                                                                                                                                                        | Calculated          |
| $IBES\ Down_i$                                                             | Indicator equal to one if IBES's rank fell by at least two positions.                                                                                                                                                                                                                                                                | Calculated          |
| $Outage_{i,q}$                                                             | Indicator equal to one if a ChatGPT incident of at least 240 minutes occurred on the calendar day immediately preceding firm $i$ 's quarter- $q$ announcement date.                                                                                                                                                                  | OpenAI              |
| $Large\ Unexpected\ Item\ (LUI_{i,q})$                                     | Indicator equal to one if firm-quarter $(i, q)$ contains a large unexpected item: any of seven Compustat non-recurring items mapped per <a href="#">Bochkay et al. (2022)</a> , with restructuring, acquisition, and special items requiring top-decile magnitude. Refines <i>Unexpected Item</i> (Panel B) with a magnitude screen. | Compustat           |
| $INFO_{i,q}$                                                               | Indicator equal to one if firm-quarter $(i, q)$ contains no large unexpected item, $1-LUI_{i,q}$ .                                                                                                                                                                                                                                   | Calculated          |

## Appendix B: Worked Examples of GenAI Aggregation of Analyst Consensus

This appendix shows the prompt and ChatGPT (gpt-4o) responses for two illustrative large-cap firms, IBM and Microsoft. Responses were obtained through the DataForSEO LLM Responses API and represent the unedited output of the model. Each firm subsection ends with a summary table mapping the cited aggregators to their upstream forecast data provider (FDP).

### B.0 Prompt

*What is the analyst consensus EPS forecast for [TICKER] for the next quarter? Please report the figures compiled by multiple third-party aggregators and cite each source.*

### B.1 International Business Machines Corporation (IBM)

#### ChatGPT (gpt-4o) response:

Here's the latest analyst consensus EPS forecast for IBM's next quarter, as compiled from several third-party aggregators.

- **MarketBeat (via Fiscal.ai):** Q2 2026 average EPS estimate is **\$2.96**, based on 2 estimates (low \$2.92, high \$2.99) ([marketbeat.com](https://marketbeat.com)).
- **TipRanks:** next quarter's earnings estimate is **\$3.02**, with a range from \$2.92 to \$3.09 ([tipranks.com](https://tipranks.com)).
- **Barchart:** next quarter average earnings estimate is **\$2.96**, based on 6 estimates (low \$2.73, high \$3.50) ([barchart.com](https://barchart.com)).
- **Yahoo Finance:** Jun 2026 average EPS estimate is **\$3.04**, based on 14 analysts (low \$2.80, high \$3.21) ([finance.yahoo.com](https://finance.yahoo.com)).

#### Summary table.

| Source        | Forecast | Upstream FDP           |
|---------------|----------|------------------------|
| MarketBeat    | \$2.96   | Independent aggregator |
| TipRanks      | \$3.02   | Independent aggregator |
| Barchart      | \$2.96   | Independent aggregator |
| Yahoo Finance | \$3.04   | S&P Capital IQ         |

### B.2 Microsoft Corporation (MSFT)

#### ChatGPT (gpt-4o) response:

Here are the analyst consensus EPS forecasts for Microsoft (MSFT) for the **next quarter**, as compiled by multiple third-party aggregators. Each figure is accompanied by its source.

- **MarketBeat:** for Q4 FY2026, the consensus EPS estimate is **\$4.04**, based on 5 analyst forecasts ([marketbeat.com](https://marketbeat.com)).
- **Zacks Investment Research:** consensus estimate for the next quarter (June 2026) is **\$4.13**, derived from 13 analyst estimates ([zacks.com](https://zacks.com)).
- **TipRanks:** next EPS estimate is **\$4.24** ([tipranks.com](https://tipranks.com)).

- **ChartMill:** consensus EPS estimate of **\$4.33** for the next earnings (Q4 FY2026), based on 39 analysts ([chartmill.com](http://chartmill.com)).
- **Seeking Alpha:** an earnings preview lists the consensus EPS at \$4.06 for Q3 FY2026 (already reported), so not applicable for the upcoming quarter ([seekingalpha.com](http://seekingalpha.com)).
- **Yahoo Finance:** for the period “Next qtr. (Jun 2026)”, average EPS estimate is **\$4.26**, based on 29 analysts (low \$3.74, high \$5.54) ([finance.yahoo.com](http://finance.yahoo.com)).
- **Investing.com:** I was unable to locate a clear next-quarter EPS consensus figure ([investing.com](http://investing.com)).

**Summary table.**

| Source        | Forecast | Upstream FDP           |
|---------------|----------|------------------------|
| MarketBeat    | \$4.04   | Independent aggregator |
| Zacks         | \$4.13   | Zacks                  |
| TipRanks      | \$4.24   | Independent aggregator |
| Yahoo Finance | \$4.26   | S&P Capital IQ         |
| ChartMill     | \$4.33   | Independent aggregator |

## Appendix C: Numerical Illustration of the Accuracy-Weighted SUE

This appendix illustrates how the accuracy-weighted SUE is constructed for a single firm-quarter. Suppose firm  $i$  in quarter  $q$  has the following backward-looking accuracy histories ( $acc_{i,q}^g = -MAFE_{i,q}^g$  over the prior four quarters, in price-scaled units) across the five FDPs (the  $z$ -scores use the sample standard deviation across the five providers), alongside their current-quarter standardized surprises  $SUE_{i,q}^g$  (in percent of price).

*Panel A. Weight construction from accuracy history.*

| Provider $g$ | $acc_{i,q}^g$ | $acc\_z_{i,q}^g$ | $\exp(acc\_z_{i,q}^g)$ | $w_{i,q}^g$ |
|--------------|---------------|------------------|------------------------|-------------|
| IBES         | -0.0010       | +1.26            | 3.54                   | 0.49        |
| FactSet      | -0.0020       | +0.63            | 1.88                   | 0.26        |
| Capital IQ   | -0.0030       | 0.00             | 1.00                   | 0.14        |
| Zacks        | -0.0040       | -0.63            | 0.53                   | 0.07        |
| Bloomberg    | -0.0050       | -1.26            | 0.28                   | 0.04        |
| Sum          |               |                  | 7.24                   | 1.00        |

I/B/E/S has the smallest absolute forecast error (largest  $acc^g$ ) and therefore the highest within-row  $z$ -score, which the softmax transform maps to the largest weight (0.49). Bloomberg has the largest absolute forecast error and receives the smallest weight (0.04). The weights sum to one and reduce to the uniform 0.20 vector when the five accuracy scores are equal.

*Panel B. Composite construction.*

| Provider $g$ | $SUE_{i,q}^g$ | $w_{i,q}^g \times SUE_{i,q}^g$ | $0.20 \times SUE_{i,q}^g$ |
|--------------|---------------|--------------------------------|---------------------------|
| IBES         | +1.20         | +0.59                          | +0.24                     |
| FactSet      | +0.80         | +0.21                          | +0.16                     |
| Capital IQ   | +1.00         | +0.14                          | +0.20                     |
| Zacks        | +1.50         | +0.11                          | +0.30                     |
| Bloomberg    | -0.50         | -0.02                          | -0.10                     |
| Total        |               | $SUE_{i,q}^{Acc} = +1.02$      | $SUE_{i,q}^{EW5} = +0.80$ |

The accuracy-weighted composite  $SUE_{i,q}^{Acc} = +1.02$  exceeds the equal-weighted composite  $SUE_{i,q}^{EW5} = +0.80$  because the two most-accurate providers in this row (I/B/E/S and FactSet) reported positive surprises, while the least-accurate provider (Bloomberg) was the lone outlier on the downside. The accuracy tilt  $SUE_{i,q}^{Acc} - SUE_{i,q}^{EW5} = +0.22$  captures the accuracy-driven correction. In the horse-race regression of Table 7, this tilt is the source of identification between the two composites: when the tilt is informative about announcement-window returns beyond the equal-weighted aggregate, the market is loading on provider-specific accuracy rather than treating all five FDPs symmetrically.